A Solution to the Accuracy/Robustness Dilemma in Impedance Control

Sang Hoon Kang, Maolin Jin, *Member, IEEE*, and Pyung Hun Chang, *Member, IEEE*

***Abstract*—It has been reported that, in impedance control, there exists a dilemma between impedance accuracy and robustness against modeling error. As a solution to this dilemma, an accurate and robust impedance control technique is developed based on internal model control structure and time-delay estimation: the former injects desired impedance and corrects modeling error, the latter estimates and compensates the nonlinear dynamics of robot manipulators. Owing to the simple structure, the proposed control is designed without requiring entire model computation or complex algorithms. In 2-DOF SCARA-type robot experiments, the accuracy and robustness of the proposed control are confirmed through comparisons with other competent controllers including impedance control with disturbance observer.**

***Index Terms*—Force control, impedance control, internal model control (IMC), time-delay control.**

I. Introduction

FOR MORE than two decades since it was first reported, impedance control has been noted and recognized as a promising unified approach to robot manipulation [1], [2]. Owing to its capability to deal with all parts of a task—free motion, constrained motion, and the transition between them—without switching between different modes of control at the moment of contact and requiring precise knowledge of environment stiffness [3], impedance control has been applied to many robotic tasks: rehabilitation [4], teleoperation [5], [6], pushing [7], prosthesis [8], and functional MRI (fMRI) [9], etc.

Recently, however, it was revealed in [10] that it is difficult to enhance robustness against modeling error without losing the accuracy at which the desired impedance is attained. This is known as the "accuracy/robustness dilemma in impedance control" [10].

The original method, suggested by Hogan [1], compensates robot dynamics by using a dynamic model [10]. In this sense, it is called dynamics-based impedance control (DB-IC) [10] and is sensitive to modeling error. Moreover DB-IC uses no additional gains for correcting the modeling error. In contrast with DB-IC, position-based impedance control (PB-IC), one of the most common implementation of impedance control [3], [10]–[12], can enhance robustness against modeling error by using an inner position control loop [10]–[12]. However, because of the modeling error, the inner loop dynamics, which is supposed to be hidden, is excited [10]. Thus, the inner loop dynamics hinders accurate realization of desired impedance dynamics.

As a tradeoff between accuracy and robustness, instantaneous-model-based impedance control (IM-IC) was proposed [10]. It also uses an inner position control loop for robustness. The difference is that the desired impedance model is replaced with an instantaneous model for accuracy. However, in this case as well, accurate realization of the desired impedance is hindered by the inner loop dynamics, unless the inner loop dynamics is identical to the desired impedance dynamics [10].

Many research works have been carried out with the aim of attaining robustness of impedance control. Notable approaches include variable structure control [13], adaptive control [14], and iterative learning control [15]. Although these approaches succeeded in enhancing robustness, they either use a robot dynamics model on real-time basis—the parameters of which are difficult to obtain and the evaluation of which is computationally demanding—or involve complex algorithms, which are not simple to put into practice. Therefore, in addition to accuracy and robustness, simplicity of the control law (minimal use of a dynamic model and a minimum number of gains) is preferable.

With these considerations in mind and in response to the given dilemma, an impedance control method was developed based on internal model control (IMC) [16] and time-delay estimation (TDE) [17].

IMC is a well-known, practical robust control method proposed by Economou and Morari [16]. It has been applied to various chemical processes [18], as well as to position control of manipulators, as an extension to mechanical systems in combination with a computed torque method [19]. IMC was chosen to inject desired impedance dynamics while correcting modeling error by using its perfect control performance. Although IMC also requires a robot dynamics model, it is easy to design the IMC loop, owing to its straightforward design strategy [16].

In order to exempt the use of a robot dynamics model, TDE [17], [20]–[22] was incorporated to estimate and compensate the dynamics with minimum information of the plant: the recent past acceleration and input force. Neither the entire dynamic model nor parameters are necessary; furthermore, one does not need to compute them at all. Consequently, TDE is simple and computationally efficient.

The disturbance observer (DOB) [23]–[25], which is noted for its robustness and accuracy as well as its simplicity and computational efficiency, appears to be based on a similar concept with TDE. A control law based on this approach is successfully used

Manuscript received August 25, 2008. First published March 4, 2009; current version published June 17, 2009. This work was supported in part by the Agency for Defense Development and in part by the Unmanned Technology Research Center (UTRC), Korea Advanced Institute of Science and Technology. Recommended by Technical Editor K. Ohnishi.

S. H. Kang and P. H. Chang are with the Robotics and Control Laboratory, Department of Mechanical Engineering, Korea Advanced Institute of Science and Technology (KAIST), Daejeon 305-701, Korea (e-mail: shkang@mecha.kaist.ac.kr).

M. Jin was with the Mechanical Engineering Research Institute, KAIST, Daejeon, Korea. He is now with the Research Institute of Industrial Science and Technology, Pohang, Korea (e-mail: mulim@rist.re.kr).

Color versions of one or more of the figures in this paper are available online at http://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/TMECH.2008.2005524

for various robot control applications including impedance control [23]–[25]. Since the similarity between TDE and DOB has been noted, it is worthwhile to clarify the differences between the impedance control with DOB and the proposed control and compare the performance of the two controllers by experiments.

Thus, this paper strives to make a threefold contribution: to propose a robust and accurate impedance control as a solution to the "accuracy/robustness dilemma in impedance control"; to show the simplicity of the proposed control law; and to demonstrate the improved robustness and accuracy of proposed control, in comparison with that of DB-IC, PB-IC, IM-IC, and the impedance control with DOB through experiments.

This paper is structured as follows. Section II briefly introduces DB-IC, PB-IC, IM-IC, and the "accuracy/robustness dilemma in impedance control." In Section III, we propose an impedance control scheme and analyze its properties. Section IV presents a comparison between the proposed approach, the DB-IC, the PB-IC, and the IM-IC through 2-DOF experiments. In Section V, differences between the proposed control and the impedance control with DOB are clarified, and the performance of the two is compared by 2-DOF experiments. Finally, in Section VI, we summarize the results and draw conclusions.

II. Accuracy/Robustness Dilemma in Impedance Control

In this section, the dilemma [10] is presented with a brief description of DB-IC, PB-IC, and IM-IC.

A. Desired Impedance

The goal of impedance control [11], [12] is to achieve desired impedance between the end-effector position and interaction force as follows:

$$\mathbf{M}_d(\ddot{\mathbf{x}}_d - \ddot{\mathbf{x}}) + \mathbf{B}_d(\dot{\mathbf{x}}_d - \dot{\mathbf{x}}) + \mathbf{K}_d(\mathbf{x}_d - \mathbf{x}) = \mathbf{F}_e \quad (1)$$

where $\mathbf{x}, \dot{\mathbf{x}}, \ddot{\mathbf{x}} \in \Re^{\mathrm{n}}$ denote the Cartesian space position, velocity, and acceleration, respectively; $\mathbf{x}_d, \dot{\mathbf{x}}_d, \ddot{\mathbf{x}}_d \in \Re^{\mathrm{n}}$ denote the Cartesian desired trajectory and its time derivatives, respectively; $\mathbf{F}_e \in \Re^{\mathrm{n}}$ denotes the interaction force on the environment exerted by the robot; and $\mathbf{M}_d, \mathbf{B}_d, \mathbf{K}_d \in \Re^{\mathrm{n}\times\mathrm{n}}$ denote the mass, damping, and stiffness of the desired impedance, respectively. In the Laplace domain, desired impedance dynamics (1) can be written as [12]

$$\mathbf{F}_e = \mathbf{Z}_d(s)\,(\mathbf{x}_d - \mathbf{x}) \quad (2)$$

with

$$\mathbf{Z}_d(s) = s^2\mathbf{M}_d + s\mathbf{B}_d + \mathbf{K}_d \quad (3)$$

where $\mathbf{Z}_d(s) \in \Re^{\mathrm{n}\times\mathrm{n}}$ denotes desired relation between $\mathbf{x}$ and $\mathbf{F}_e$.

B. Robot Dynamics

For the Cartesian space control, usually two forms of robot dynamics are considered: one form [10] is

$$\boldsymbol{\tau}_u = \mathbf{M}_\theta(\boldsymbol{\theta})\mathbf{J}^{-1}(\boldsymbol{\theta})(\ddot{\mathbf{x}} - \dot{\mathbf{J}}\dot{\boldsymbol{\theta}}) + \mathbf{N}_\theta(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}}) + \mathbf{J}^T(\boldsymbol{\theta})\mathbf{F}_e \quad (4)$$

where $\boldsymbol{\theta}, \dot{\boldsymbol{\theta}} \in \Re^{\mathrm{n}}$ denote the joint position and velocity, respectively; $\mathbf{M}_\theta(\boldsymbol{\theta}) \in \Re^{\mathrm{n}\times\mathrm{n}}$ denotes the inertia matrix; $\mathbf{N}_\theta(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}}) \in \Re^{\mathrm{n}}$ denote the centrifugal, Coriolis, gravitational, and friction forces; $\mathbf{J}(\boldsymbol{\theta}) \in \Re^{\mathrm{n}\times\mathrm{n}}$ denotes the Jacobian; and $\boldsymbol{\tau}_u \in \Re^{\mathrm{n}}$ denotes the input torque.

The other form [26] is easily obtained from (4) with the static relation $\boldsymbol{\tau}_u = \mathbf{J}^T\mathbf{F}_u$, where $\mathbf{F}_u \in \Re^{\mathrm{n}}$ denotes the Cartesian space input force

$$\mathbf{F}_u = \mathbf{M}_x(\boldsymbol{\theta})\ddot{\mathbf{x}} + \mathbf{N}_x(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}}) + \mathbf{F}_e \quad (5)$$

where $\mathbf{M}_x = \mathbf{J}^{-T}\mathbf{M}_\theta\mathbf{J}^{-1}$ and $\mathbf{N}_x = \mathbf{J}^{-T}[\mathbf{N}_\theta - \mathbf{M}_\theta\mathbf{J}^{-1}\dot{\mathbf{J}}\dot{\boldsymbol{\theta}}]$.

C. Dynamics-Based Impedance Control

Based on (4), the following DB-IC law is derived to achieve the desired impedance dynamics of (1):

$$\boldsymbol{\tau}_u = \hat{\mathbf{M}}_\theta(\boldsymbol{\theta})\mathbf{J}^{-1}(\boldsymbol{\theta})(\mathbf{u}_d - \dot{\mathbf{J}}\dot{\boldsymbol{\theta}}) + \hat{\mathbf{N}}_\theta(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}}) + \mathbf{J}^T(\boldsymbol{\theta})\mathbf{F}_e \quad (6)$$

with

$$\mathbf{u}_d = \ddot{\mathbf{x}}_d + \mathbf{M}_d^{-1}[\mathbf{B}_d(\dot{\mathbf{x}}_d - \dot{\mathbf{x}}) + \mathbf{K}_d(\mathbf{x}_d - \mathbf{x}) - \mathbf{F}_e] \quad (7)$$

where $\hat{\mathbf{M}}_\theta(\boldsymbol{\theta}) \in \Re^{\mathrm{n}\times\mathrm{n}}$ and $\hat{\mathbf{N}}_\theta(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}}) \in \Re^{\mathrm{n}}$ denote estimates of $\mathbf{M}_\theta$ and $\mathbf{N}_\theta$, respectively; and $\mathbf{u}_d \in \Re^{\mathrm{n}}$ denotes the new control input. In the Laplace domain, $\mathbf{u}_d$ can be written as follows:

$$\mathbf{u}_d = \mathbf{M}_d^{-1}\mathbf{Z}_d(s)\mathbf{x}_d - \mathbf{M}_d^{-1}\mathbf{F}_e - \mathbf{M}_d^{-1}(s\mathbf{B}_d + \mathbf{K}_d)\mathbf{x}. \quad (8)$$

Clearly, the DB-IC in (6) uses dynamics-based compensation. Thus, robustness of the DB-IC entirely relies on the estimated values, $\hat{\mathbf{M}}_\theta$ and $\hat{\mathbf{N}}_\theta$.

D. Position-Based Impedance Control

From [10]–[12], one may summarize PB-IC as follows. The desired impedance model modifies the desired trajectory by using the measured interaction force. The modified trajectory is then imposed as a command of the inner position control loop [10]–[12]. The position command of the inner loop is the solution to the desired impedance model:

$$\mathbf{M}_d(\ddot{\mathbf{x}}_d - \ddot{\mathbf{x}}_r) + \mathbf{B}_d(\dot{\mathbf{x}}_d - \dot{\mathbf{x}}_r) + \mathbf{K}_d(\mathbf{x}_d - \mathbf{x}_r) = \mathbf{F}_e \quad (9)$$

where $\mathbf{x}_r, \dot{\mathbf{x}}_r, \ddot{\mathbf{x}}_r \in \Re^{\mathrm{n}}$ denote the inner loop position command and its time derivatives, respectively. In the Laplace domain, the inner loop command is computed as follows:

$$\begin{aligned}\mathbf{x}_r &= \mathbf{Z}_d^{-1}(s)\mathbf{M}_d[\mathbf{M}_d^{-1}\mathbf{Z}_d(s)\mathbf{x}_d - \mathbf{M}_d^{-1}\mathbf{F}_e] \\ &= \mathbf{x}_d - \mathbf{Z}_d^{-1}(s)\mathbf{F}_e.\end{aligned} \quad (10)$$

If the inner position control loop guarantees perfect tracking performance ($\mathbf{x} = \mathbf{x}_r$), then from (9), desired impedance is attained perfectly [12]. Note that that if $\mathbf{F}_e$ is zero, $\mathbf{x}_r$ becomes $\mathbf{x}_d$. Thus, in free space, small deviation between $\mathbf{x}_r$ and $\mathbf{x}$ implies small position tracking error [12].

Any position control law can be used for the inner loop. In this study, an experimentally verified method presented in [11] is selected. It also uses the same dynamics compensation of (6). The difference is that $\mathbf{u}_d$ is replaced with $\mathbf{u}_p$

$$\mathbf{u}_p = \ddot{\mathbf{x}}_r + \mathbf{K}_v(\dot{\mathbf{x}}_r - \dot{\mathbf{x}}) + \mathbf{K}_p(\mathbf{x}_r - \mathbf{x}) \quad (11)$$

where $\mathbf{K}_p, \mathbf{K}_v \in \Re^{\mathrm{n}\times\mathrm{n}}$ denote the gains of the decoupled proportional and derivative (PD) controller, which can be designed independently of the desired impedance parameters.

E. Accuracy/Robustness Dilemma in Impedance Control

Originally, for simplicity, the dilemma was analyzed by using a linear 1-DOF robot [10]. In this study, considering a real robot system, the same dilemma is analyzed by using a nonlinear multi-DOF robot.

The robot dynamics (4) can be rewritten as follows:

$$\begin{aligned}\boldsymbol{\tau}_u &= \hat{\mathbf{M}}_\theta \mathbf{J}^{-1}(\ddot{\mathbf{x}} - \dot{\mathbf{J}}\dot{\boldsymbol{\theta}}) + \hat{\mathbf{N}}_\theta + \mathbf{J}^T\mathbf{F}_e \\ &\quad + (\mathbf{M}_\theta - \hat{\mathbf{M}}_\theta)\mathbf{J}^{-1}(\ddot{\mathbf{x}} - \dot{\mathbf{J}}\dot{\boldsymbol{\theta}}) + (\mathbf{N}_\theta - \hat{\mathbf{N}}_\theta). \quad (12)\end{aligned}$$

Applying DB-IC in (6) to the robot dynamics (12) leads to

$$\hat{\mathbf{M}}_\theta \mathbf{J}^{-1}(\mathbf{u} - \ddot{\mathbf{x}}) = (\mathbf{M}_\theta - \hat{\mathbf{M}}_\theta)\mathbf{J}^{-1}(\ddot{\mathbf{x}} - \dot{\mathbf{J}}\dot{\boldsymbol{\theta}}) + (\mathbf{N}_\theta - \hat{\mathbf{N}}_\theta). \quad (13)$$

Note that (13) representing a relationship common to *both* DB-IC and PB-IC, $\mathbf{u}$ represents either $\mathbf{u}_d$ or $\mathbf{u}_p$.

Now, multiplying $(\hat{\mathbf{M}}_\theta \mathbf{J}^{-1})^{-1}$ on both sides of (13) and terming the right-hand side as *dynamics estimation error*, $\boldsymbol{\eta}_{(t)} \in \Re^n$, we have the following definition:

$$\boldsymbol{\eta}_{(t)} \triangleq \mathbf{J}\hat{\mathbf{M}}_\theta^{-1}[(\mathbf{M}_\theta - \hat{\mathbf{M}}_\theta)\mathbf{J}^{-1}(\ddot{\mathbf{x}} - \dot{\mathbf{J}}\dot{\boldsymbol{\theta}}) + (\mathbf{N}_\theta - \hat{\mathbf{N}}_\theta)]. \quad (14)$$

From (14), $\boldsymbol{\eta}$, in the form of acceleration, is a difference between real and estimated dynamics. From (13) and (14), it follows that

$$\boldsymbol{\eta}_{(t)} = \mathbf{u}_{(t)} - \ddot{\mathbf{x}}_{(t)} \quad (15)$$

which will be used later. Note that, from (14) and (15), $\boldsymbol{\eta}$ can be regarded as uncompensated coupled dynamics, which couples motions of several axes. From (15), it may be interpreted that $\mathbf{u}$ and $\mathbf{x}$ have a linear second-order dynamics subject to dynamics estimation error $\boldsymbol{\eta}$ stemming from modeling error, and is expressed as follows:

$$\mathbf{u} = s^2\mathbf{x} + \boldsymbol{\eta}. \quad (16)$$

1) Accuracy and Robustness of the DB-IC: The new control input $\mathbf{u}_d$ in (8) is arranged into two parts as follows:

$$\mathbf{u}_d = \mathbf{v}_d - \mathbf{M}_d^{-1}(s\mathbf{B}_d + \mathbf{K}_d)\mathbf{x} \quad (17)$$

where $\mathbf{v}_d \in \Re^n$ is defined as

$$\mathbf{v}_d \triangleq \mathbf{M}_d^{-1}\mathbf{Z}_d(s)\mathbf{x}_d - \mathbf{M}_d^{-1}\mathbf{F}_e. \quad (18)$$

Note that $\mathbf{v}_d$ represents the combination of the reference input and interaction force, and the other part stands for the position feedback input. Substituting (17) into (16)—$\mathbf{u}$ of (16) for DB-IC is $\mathbf{u}_d$—and solving for $\mathbf{x}$, we have

$$\mathbf{x} = \mathbf{Z}_d^{-1}(s)\mathbf{M}_d(\mathbf{v}_d - \boldsymbol{\eta}) \quad (19)$$

which is illustrated in Fig. 1(a). From (10) and (18), $\mathbf{v}_d$ can be rewritten as follows:

$$\mathbf{v}_d = \mathbf{M}_d^{-1}\mathbf{Z}_d(s)\mathbf{x}_r. \quad (20)$$

Substituting (20) into (19) and solving for $\mathbf{x}_r - \mathbf{x}$, we have

$$\begin{aligned}\mathbf{x}_r - \mathbf{x} &= \mathbf{Z}_d^{-1}(s)\mathbf{M}_d\boldsymbol{\eta} \\ &= (s^2\mathbf{I} + s\mathbf{M}_d^{-1}\mathbf{B}_d + \mathbf{M}_d^{-1}\mathbf{K}_d)^{-1}\boldsymbol{\eta}. \quad (21)\end{aligned}$$

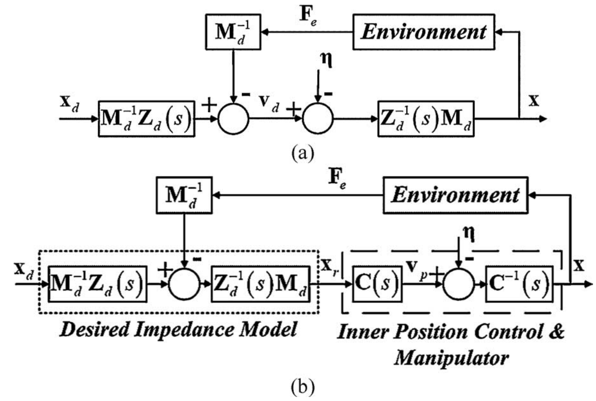

Fig. 1. Block diagram of the DB-IC and PB-IC. (a) DB-IC: desired dynamics is disturbed by dynamics estimation error $\boldsymbol{\eta}$. (b) PB-IC: dynamics estimation error $\boldsymbol{\eta}$ excites inner loop dynamics $\mathbf{C}^{-1}(s)$.

Accurate realization of desired impedance is disturbed by $\boldsymbol{\eta}$ stemming from modeling error. Moreover, except for the parameters to achieve the desired impedance, there are no adjustable parameters to reduce the effect of $\boldsymbol{\eta}$. Thus, DB-IC is vulnerable and sensitive to the modeling error. Especially when small stiffness and large mass are desired, the sensitivity is high [10].

2) Accuracy and Robustness of the PB-IC: Similar to (17), $\mathbf{u}_p$ in (11) can be divided into two parts as follows:

$$\mathbf{u}_p = \mathbf{v}_p - (s\mathbf{K}_v + \mathbf{K}_p)\mathbf{x} \quad (22)$$

where $\mathbf{v}_p \in \Re^n$ is defined as

$$\mathbf{v}_p \triangleq \mathbf{C}(s)\mathbf{x}_r \quad (23)$$

with $\mathbf{C}(s) = s^2\mathbf{I} + s\mathbf{K}_v + \mathbf{K}_p$. Clearly, the first part represents the reference input and the second feedback input. Substituting (22) into (16)—$\mathbf{u}$ of (16) for PB-IC is $\mathbf{u}_p$—and solving for $\mathbf{x}$, we have

$$\mathbf{x} = \mathbf{C}^{-1}(s)(\mathbf{v}_p - \boldsymbol{\eta}). \quad (24)$$

The relations (10) and (24), are illustrated in Fig. 1(b). By substituting (23) into (24) and solving for $\mathbf{x}_r - \mathbf{x}$, we have

$$\begin{aligned}\mathbf{x}_r - \mathbf{x} &= \mathbf{C}^{-1}(s)\boldsymbol{\eta} \\ &= (s^2\mathbf{I} + s\mathbf{K}_v + \mathbf{K}_p)^{-1}\boldsymbol{\eta}. \quad (25)\end{aligned}$$

Comparison of (21) and (25) displays that $\mathbf{C}^{-1}(s)$ contains free gains $\mathbf{K}_p$ and $\mathbf{K}_v$, and that increasing them attenuates the effect of $\boldsymbol{\eta}$. Thus, PB-IC has better robustness compared with that of DB-IC. However, it is clear from Fig. 1(b) that the dynamics $\mathbf{C}^{-1}(s)$, which is meant to be canceled by $\mathbf{C}(s)$, is excited by $\boldsymbol{\eta}$ and hinders accurate realization of the desired impedance. Thus, the inaccuracy is related to the inner loop dynamics $\mathbf{C}^{-1}(s)$. In the worst case, the robot loses contact and oscillates [10].

3) Accuracy/Robustness Dilemma in Impedance Control: Although DB-IC can realize desired impedance accurately with precise modeling, it is sensitive to the modeling error. PB-IC enhances robustness against the modeling error. However, PB-IC realizes desired impedance inaccurately, and this inaccuracy is related to its inherent inner loop dynamics, which is, in general,

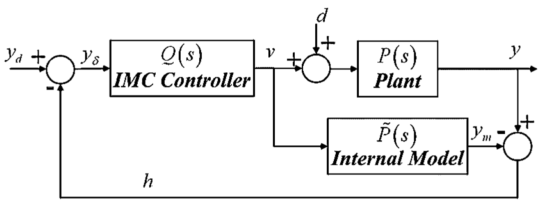

Fig. 2. Block diagram of the IMC, where Q denotes the IMC controller, P signifies the plant, and $\tilde{P}$ denotes the internal model.

different from the desired impedance dynamics. Thus, there is a dilemma between accuracy and robustness [10].

4) Instantaneous-Model-Based Impedance Control: As a tradeoff between robustness and accuracy, IM-IC has been proposed. Instead of the desired impedance model in (9), IM-IC uses an instantaneous impedance model [10] as follows:

$$\mathbf{M}_d(\ddot{\mathbf{x}}_d - \ddot{\mathbf{x}}_{im}) + \mathbf{B}_d(\dot{\mathbf{x}}_d - \dot{\mathbf{x}}_{im}) + \mathbf{K}_d(\mathbf{x}_d - \mathbf{x}_{im}) = \mathbf{F}_e \tag{26}$$

where $\mathbf{x}_{im}, \dot{\mathbf{x}}_{im}, \ddot{\mathbf{x}}_{im} \in \Re^{\mathrm{n}}$ denote the inner loop position command and its time derivatives, respectively. The next step velocity command is predicted as follows [10]:

$$\begin{aligned}\dot{\mathbf{x}}_{im(t+L)} &= \dot{\mathbf{x}}_{im(t)} + \int_t^{t+L} \{\ddot{\mathbf{x}}_{d(t)} + \mathbf{M}_d^{-1}[\mathbf{B}_d(\dot{\mathbf{x}}_{d(t)} - \dot{\mathbf{x}}_{im(\lambda)}) \\ &\quad + \mathbf{K}_d(\mathbf{x}_{d(t)} - \mathbf{x}_{im(\lambda)}) - \mathbf{F}_e]\}d\lambda \\ \dot{\mathbf{x}}_{im(t)} &= \dot{\mathbf{x}}_{(t)}, \quad \mathbf{x}_{im(t)} = \mathbf{x}_{(t)}.\end{aligned} \tag{27}$$

Similarly, $\mathbf{x}_{im(t+L)}$ is computed by double integration [10]. If $\ddot{\mathbf{x}}_{im(t+L)}$ is needed, then it can be computed as below [10]

$$\begin{aligned}\ddot{\mathbf{x}}_{im(t+L)} &= \ddot{\mathbf{x}}_d + \mathbf{M}_d^{-1}[\mathbf{B}_d(\dot{\mathbf{x}}_d - \dot{\mathbf{x}}_{im(t+L)}) \\ &\quad + \mathbf{K}_d(\mathbf{x}_d - \mathbf{x}_{im(t+L)}) - \mathbf{F}_e].\end{aligned} \tag{28}$$

The instantaneous model ensures that the desired impedance dynamics is maintained in the actual position of the robot, instead of the predicted position in (10). However, in this case as well, accurate realization of desired impedance is hindered by the inner loop dynamics [10].

III. Proposed Impedance Control

A. Internal Model Control

IMC, proposed by Economou and Morari [16], is shown in Fig. 2. The closed-loop dynamics due to IMC is given as

$$y = PQ[1+Q(P-\tilde{P})]^{-1}y_d + (1-\tilde{P}Q)[1+Q(P-\tilde{P})]^{-1}Pd. \tag{29}$$

From (29), one can easily derive the so-called *perfect control* property: assuming that $Q = \tilde{P}^{-1}$ and the closed-loop system is stable, then perfect control ($y = y_d$) can be achieved. Note that the condition $Q = \tilde{P}^{-1}$ only restricts the relation between the IMC controller and internal model, not the plant. In addition, the design is straightforward: the internal model is set to be the same as the plant model, and the IMC controller is the inverse of the internal model [16].

In short, IMC has the following properties: *perfect control* performance and a straightforward design strategy. While the perfect control property is promising to resolving the dilemma, the selection of the internal model requires a robot dynamics model on real-time basis, sharing the same difficulty with the other control laws mentioned in Section I and necessitating an effective scheme such as the TDE.

B. Time-Delay Estimation

TDE is a model-independent simple dynamics estimation technique [17]. The dynamics of (5) can be divided into two terms, the known term and the unknown and uncertain nonlinear terms represented by $\mathbf{H}(\boldsymbol{\theta}, \mathbf{x}, \dot{\mathbf{x}}, \ddot{\mathbf{x}}) \in \Re^{\mathrm{n}}$ as follows:

$$\mathbf{F}_u = \bar{\mathbf{M}}_x\ddot{\mathbf{x}} + \mathbf{H}(\boldsymbol{\theta}, \mathbf{x}, \dot{\mathbf{x}}, \ddot{\mathbf{x}}) \tag{30}$$

where $\bar{\mathbf{M}}_x \in \Re^{\mathrm{n}\times\mathrm{n}}$ denotes a matrix representing the known part of $\mathbf{M}_x$, and $\mathbf{H}(\boldsymbol{\theta}, \mathbf{x}, \dot{\mathbf{x}}, \ddot{\mathbf{x}}) \triangleq [\mathbf{M}_x(\boldsymbol{\theta}) - \bar{\mathbf{M}}_x]\ddot{\mathbf{x}} + \mathbf{N}_x(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}}) + \mathbf{F}_e$. Assume that $\mathbf{H}(\boldsymbol{\theta}, \mathbf{x}, \dot{\mathbf{x}}, \ddot{\mathbf{x}})$ be continuous or at least piecewise continuous, and let $\mathbf{H}_{(t)}$ denote $\mathbf{H}(\boldsymbol{\theta}, \mathbf{x}, \dot{\mathbf{x}}, \ddot{\mathbf{x}})$ at time t. Then, for a sufficiently small time delay L, $\hat{\mathbf{H}}_{(t)}$, the time-delayed estimate of $\mathbf{H}_{(t)}$ can be obtained as below [17]

$$\hat{\mathbf{H}}_{(t)} = \mathbf{H}_{(t-L)} = \mathbf{F}_{u(t-L)} - \bar{\mathbf{M}}_x\ddot{\mathbf{x}}_{(t-L)}. \tag{31}$$

Normally, L is set to be the sampling period in digital implementation. TDE does not require a dynamic model or its parameters except for $\bar{\mathbf{M}}_x$. It only needs the recent past information of acceleration and input force.

Similar to the definition of the dynamics estimation error in (14), TDE error, $\boldsymbol{\eta}_{T(t)} \in \Re^{\mathrm{n}}$, is defined as follows:

$$\boldsymbol{\eta}_{T(t)} \triangleq \bar{\mathbf{M}}_x^{-1}[\mathbf{H}_{(t)} - \hat{\mathbf{H}}_{(t)}]. \tag{32}$$

From (32), similar to $\boldsymbol{\eta}$, TDE error $\boldsymbol{\eta}_T$, in the form of acceleration, can also be considered as a difference between real and estimated dynamics. That $\boldsymbol{\eta}_T$ too is a dynamics estimation error will be shown in (43).

TDE is applied to the proposed control to remove the need of an entire robot dynamics model.

C. Derivation of the Proposed Impedance Control Law

1) IMC Structure for Impedance Control: Comparison of the structure of DB-IC in Fig. 1(a) with that of IMC in Fig. 2 shows that DB-IC is an IMC applicable form: the feedforward controller $\mathbf{M}_d^{-1}\mathbf{Z}_d$ and the plant $\mathbf{Z}_d^{-1}\mathbf{M}_d$, with $\boldsymbol{\eta}$ acting as an input disturbance to the plant. The only difference is that the interaction force $\mathbf{F}_e$ is separated from $\boldsymbol{\eta}$ for impedance control. Thus, the *perfect control* property of IMC should be able to improve the robustness against modeling error and the impedance accuracy regardless of environment stiffness—this is the *key concept* of this paper.

By following the standard IMC design procedure, a parallel internal model loop is added to the closed loop in Fig. 1(a). The internal model $\tilde{\mathbf{P}}$ is simply selected as follows:

$$\tilde{\mathbf{P}} = \mathbf{Z}_d^{-1}(s)\mathbf{M}_d. \tag{33}$$

$\tilde{\mathbf{P}}$ is invertible; therefore, the IMC controller $\mathbf{Q}$ is determined as

$$\mathbf{Q} = \mathbf{M}_d^{-1}\mathbf{Z}_d(s) \tag{34}$$

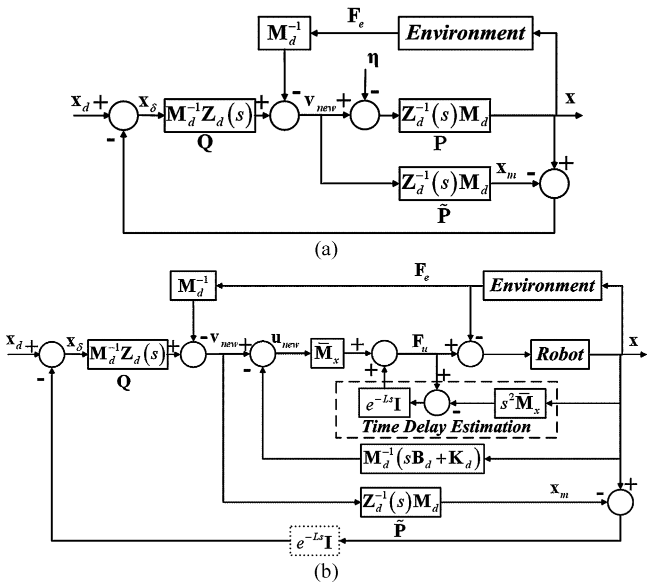

Fig. 3. Block diagrams of the proposed control. (a) Proposed IMC structure for impedance control: instead of introducing any other dynamics (such as $\mathbf{C}(s)$ of PB-IC), this structure can compensate dynamics estimation error $\boldsymbol{\eta}$ by using parallel internal model, which is the same as desired impedance model. (b) Expanded block diagram of the proposed control.

which is already given in Fig. 1(a). Adding IMC feedback, we can achieve the proposed control as illustrated in Fig. 3(a). The input to $\mathbf{Q}$ is changed from $\mathbf{x}_d$ to $\mathbf{x}_\delta$, and thus, $\mathbf{v}_d$ is replaced with $\mathbf{v}_{\mathrm{new}}$

$$\mathbf{v}_{\mathrm{new}} = \mathbf{M}_d^{-1}\mathbf{Z}_d(s)\mathbf{x}_\delta - \mathbf{M}_d^{-1}\mathbf{F}_e. \tag{35}$$

Accordingly, $\mathbf{u}_d$ in (17) is replaced with $\mathbf{u}_{\mathrm{new}}$

$$\mathbf{u}_{\mathrm{new}} = \mathbf{v}_{\mathrm{new}} - \mathbf{M}_d^{-1}(s\mathbf{B}_d + \mathbf{K}_d)\mathbf{x}. \tag{36}$$

From Fig. 3(a), one can directly deduce the following relationships:

$$\mathbf{x} = \mathbf{Z}_d^{-1}(s)\mathbf{M}_d(\mathbf{v}_{\mathrm{new}} - \boldsymbol{\eta}) \tag{37}$$

$$\mathbf{x} - \mathbf{x}_m = -\mathbf{Z}_d^{-1}(s)\mathbf{M}_d\boldsymbol{\eta} \tag{38}$$

$$\begin{aligned}\mathbf{v}_{\mathrm{new}} &= \mathbf{M}_d^{-1}\mathbf{Z}_d(s)[\mathbf{x}_d - (\mathbf{x} - \mathbf{x}_m)] - \mathbf{M}_d^{-1}\mathbf{F}_e \\ &= \mathbf{v}_d + \underbrace{\boldsymbol{\eta}}_{\text{correcting dynamics estimation error}}.\end{aligned} \tag{39}$$

From (38), the additional IMC feedback term $\mathbf{x} - \mathbf{x}_m$ is a function of $\boldsymbol{\eta}$ only. Thus, as shown in (39), the additional IMC feedback is a dynamics estimation error correction term. Substituting (20) and (39) into (37), and solving for $\mathbf{x}_r - \mathbf{x}$, we have

$$\mathbf{x}_r - \mathbf{x} = \mathbf{0}_n \tag{40}$$

where $\mathbf{0}_n \in \mathfrak{R}^{\mathrm{n}}$ denotes a zero vector. Desired impedance is perfectly realized regardless of dynamics estimation error. Therefore, accuracy and robustness are achieved with the proposed IMC structure. In this sense, the proposed IMC structure is a perfect solution to the dilemma. In addition, the design is accomplished in a straightforward manner. Further, this structure requires only the desired impedance parameters.

However, because of the required computation of the entire robot dynamics, the simplicity factor needs to be improved.

2) Model-Independent Dynamics Compensation: For simplicity, the complex-model-based compensation (6) is replaced with TDE. From (30) and (31), the following dynamics compensation is obtained [17]:

$$\begin{cases}\mathbf{F}_{u(t)} = \bar{\mathbf{M}}_x\mathbf{u}_{\mathrm{new}(t)} + \mathbf{F}_{u(t-L)} - \bar{\mathbf{M}}_x\ddot{\mathbf{x}}_{(t-L)} \\ \boldsymbol{\tau}_{u(t)} = \mathbf{J}^T\mathbf{F}_{u(t)}.\end{cases} \tag{41}$$

Subtraction of (30) from (41), and a simple mathematical manipulation, yields

$$\mathbf{u}_{\mathrm{new}(t)} - \ddot{\mathbf{x}}_{(t)} = \bar{\mathbf{M}}_x^{-1}[\mathbf{H}_{(t)} - \hat{\mathbf{H}}_{(t)}] = \boldsymbol{\eta}_{T(t)}. \tag{42}$$

From (42), it follows that

$$\boldsymbol{\eta}_{T(t)} = \mathbf{u}_{\mathrm{new}(t)} - \ddot{\mathbf{x}}_{(t)}. \tag{43}$$

By comparing (43) with (15), it can be found that $\mathbf{u}_{\mathrm{new}}$ and $\boldsymbol{\eta}_T$ are matched with $\mathbf{u}$ and $\boldsymbol{\eta}$, respectively. Again, we can confirm that TDE error $\boldsymbol{\eta}_T$ is also a dynamics estimation error, which couples motions of several axes. In the Laplace domain, (43) becomes

$$\mathbf{u}_{\mathrm{new}} = s^2\mathbf{x} + \boldsymbol{\eta}_T. \tag{44}$$

In comparison with (16), the only difference is that $\boldsymbol{\eta}$ is replaced by $\boldsymbol{\eta}_T$, which can be corrected by the proposed IMC structure.

3) Proposed Impedance Control: From (35), (36), and (41), the proposed control law is obtained as follows:

$$\begin{cases}\mathbf{F}_{u(t)} = \bar{\mathbf{M}}_x\mathbf{u}_{\mathrm{new}(t)} + \mathbf{F}_{u(t-L)} - \bar{\mathbf{M}}_x\ddot{\mathbf{x}}_{(t-L)} \\ \boldsymbol{\tau}_{u(t)} = \mathbf{J}^T\mathbf{F}_{u(t)}\end{cases} \tag{45}$$

where

$$\mathbf{u}_{\mathrm{new}(t)} = \mathbf{v}_{\mathrm{new}(t)} - \mathbf{M}_d^{-1}(\mathbf{B}_d\dot{\mathbf{x}}_{(t)} + \mathbf{K}_d\mathbf{x}_{(t)}) \tag{46}$$

with

$$\mathbf{v}_{\mathrm{new}(t)} = \ddot{\mathbf{x}}_{\delta(t)} + \mathbf{M}_d^{-1}(\mathbf{B}_d\dot{\mathbf{x}}_{\delta(t)} + \mathbf{K}_d\mathbf{x}_{\delta(t)} - \mathbf{F}_{e(t)}) \tag{47}$$

and

$$\mathbf{x}_{\delta(t)} = \mathbf{x}_{d(t)} - \mathbf{x}_{(t-L)} + \mathbf{x}_{m(t-L)}. \tag{48}$$

An expanded block diagram of the closed-loop system due to the proposed control is shown in Fig. 3(b).[1]

D. Discussion of the Proposed Impedance Control Law

1) Accuracy and Robustness: Substituting (47) and (48) into (46) and subtracting (7) gives the control input of the proposed

[1]The feedback time delay L is explicitly included in IMC feedback considering discrete-time domain implementation. See $\mathbf{x}_{(t-L)}$ and $\mathbf{x}_{m(t-L)}$ in (48), and the dotted block in Fig. 3(b). Feedback time delay is unavoidable for discrete-time domain implementation.

control as follows:

$$\begin{aligned}\mathbf{u}_{\text{new}(t)} = \mathbf{u}_{d(t)} - \{(\ddot{\mathbf{x}}_{(t-L)} - \ddot{\mathbf{x}}_{m(t-L)}) \\ + \mathbf{M}_d^{-1}[\mathbf{B}_d(\dot{\mathbf{x}}_{(t-L)} - \dot{\mathbf{x}}_{m(t-L)}) \\ + \mathbf{K}_d(\mathbf{x}_{(t-L)} - \mathbf{x}_{m(t-L)})]\}.\end{aligned} \tag{49}$$

In the Laplace domain, (49) becomes

$$\mathbf{u}_{\text{new}} = \mathbf{u}_d - \mathbf{M}_d^{-1}\mathbf{Z}_d(s)e^{-Ls}(\mathbf{x} - \mathbf{x}_m). \tag{50}$$

Similar to the derivation of (38), in this case, $\mathbf{x} - \mathbf{x}_m$ becomes

$$\mathbf{x} - \mathbf{x}_m = -\mathbf{Z}_d^{-1}(s)\mathbf{M}_d\boldsymbol{\eta}_T. \tag{51}$$

The only difference between (38) and (51) is that $\boldsymbol{\eta}$ is replaced with $\boldsymbol{\eta}_T$. In the time domain, from (50) and (51), $\mathbf{u}_{\text{new}}$ can be written as

$$\mathbf{u}_{\text{new}(t)} = \mathbf{u}_{d(t)} + \boldsymbol{\eta}_{T(t-L)}. \tag{52}$$

Substituting (8), (50), and (51) into (44), and solving for $\mathbf{x}_r - \mathbf{x}$, we have

$$\begin{aligned}\mathbf{x}_r - \mathbf{x} &= \mathbf{Z}_d^{-1}(s)\mathbf{M}_d(1 - e^{-Ls})\boldsymbol{\eta}_T \\ &= (s^2\mathbf{I} + s\mathbf{M}_d^{-1}\mathbf{B}_d + \mathbf{M}_d^{-1}\mathbf{K}_d)^{-1}(1 - e^{-Ls})\boldsymbol{\eta}_T.\end{aligned} \tag{53}$$

Comparison of (53) with both (21) and (25) reveals that the dynamics estimation error (in this case, TDE error $\boldsymbol{\eta}_{T(t)}$) is corrected by a step-delayed term $\boldsymbol{\eta}_{T(t-L)}$ without introducing any other impedance such as $\mathbf{C}(s)$ of the PB-IC. Further, if the IMC feedback has no time delay, $\mathbf{x}$ becomes identical to $\mathbf{x}_r$, i.e., desired impedance dynamics can be perfectly realized.

From the control theory point of view, the proposed control is a 2-DOF control, which has a noteworthy feature that the command input response and the closed-loop characteristics can be designed independently [27], [28]. If $\boldsymbol{\eta}_T = \mathbf{0}_n$, then the IMC feedback becomes zero, and the command input response hence coincides with the desired response (i.e., $\mathbf{x} = \mathbf{x}_r$) regardless of e^{-Ls}. If $\boldsymbol{\eta}_T \neq \mathbf{0}_n$, the IMC feedback term is nonzero but is forced to converge to zero through the feedback loop with e^{-Ls}. This explains why the proposed control can enhance robustness and accuracy without conflict. The following comparison can give us more clear view on this point.

For a quantitative comparison of DB-IC, PB-IC, and proposed control, we assume that: 1) $\mathbf{M}_d, \mathbf{B}_d, \mathbf{K}_d, \mathbf{K}_p$, and $\mathbf{K}_v$ are diagonal matrices and 2) each element of $\boldsymbol{\eta}$ and $\boldsymbol{\eta}_T$ have the same magnitude, i.e., $|\eta_i| = |\eta_{Ti}|$, where $\eta_i, \eta_{Ti} \in \Re$ denote the ith element of $\boldsymbol{\eta}$ and $\boldsymbol{\eta}_T$, respectively. With these assumptions, the ith row equation of (21), (25), and (53) can be written as follows:

$$x_{ri} - x_i = \left[\frac{1}{(s^2 + sM_{di}^{-1}B_{di} + M_{di}^{-1}K_{di})}\right]\eta_i \tag{54}$$

$$x_{ri} - x_i = \left[\frac{1}{(s^2 + sK_{vi} + K_{pi})}\right]\eta_i \tag{55}$$

and

$$x_{ri} - x_i = \left[\frac{(1 - e^{-Ls})}{(s^2 + sM_{di}^{-1}B_{di} + M_{di}^{-1}K_{di})}\right]\eta_{Ti} \tag{56}$$

where x_{ri} and x_i denote the ith element of $\mathbf{x}_r$ and $\mathbf{x}$, respectively. Equation (54) is that of DB-IC, (55) is that of PB-IC, and (56) is that of proposed control. Thus, $|x_{ri} - x_i|$ of each control

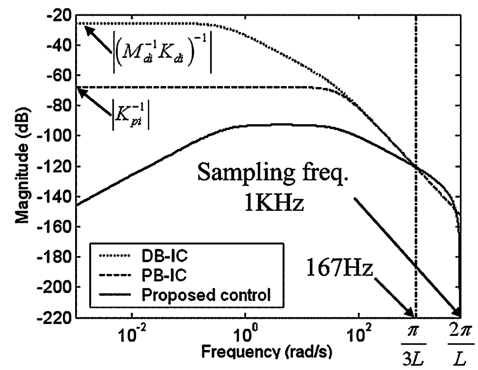

Fig. 4. Magnitude plot of transfer functions of the DB-IC, the PB-IC, and the proposed control between $x_{ri} - x_i$ and dynamics estimation error: proposed control most effectively reduces the effect of dynamics estimation error.

method entirely relies on the magnitude of transfer functions between $x_{ri} - x_i$ and dynamics estimation error (η_i or η_{Ti}). The magnitude plot of transfer functions is shown in Fig. 4. For this plot, the following parameters are used: $L = 1$ ms, $M_{di} = 20$ kg, $B_{di} = 900$ N·s/m, $K_{di} = 400$ N/m, $K_{pi} = 2500$, and $K_{vi} = 90$. These values are used for experiments in Section IV. Proposed control shows smallest magnitude in a wide range of frequency: from 0 to 167 Hz. Therefore, accurate realization of desired impedance and good position tracking performance are expected. From this comparison, clearly, one can see that the term e^{-Ls} is related to the robustness against $\boldsymbol{\eta}$, and that, due to e^{-Ls}, $\boldsymbol{\eta}$ can be effectively corrected.

2) Simplicity: The IMC loop requires no other gains except the desired impedance parameters, and can be designed in a straightforward manner. The TDE needs one gain, $\bar{\mathbf{M}}_x$, and only the information of one-step-delayed acceleration and input force instead of computation of complex nonlinear dynamics such as Coriolis and centrifugal force, gravity, and friction force, thus providing a simple approach. Overall, the proposed control fully shares the advantages of both IMC and TDE.

E. Practical Considerations

1) Removal of Internal Model Computation: The proposed control law needs $\mathbf{x}_m$, a solution to the following differential equation:

$$\ddot{\mathbf{x}}_{m(t)} + \mathbf{M}_d^{-1}\left(\mathbf{B}_d\dot{\mathbf{x}}_{m(t)} + \mathbf{K}_d\mathbf{x}_{m(t)}\right) = \mathbf{v}_{\text{new}(t)}. \tag{57}$$

To calculate $\mathbf{x}_m$, one has to solve the previous differential equation in real time. This might degrade the simplicity of the proposed control. Thus, improvement is needed here.

One typical method is equating $\mathbf{x}_m$ with $\mathbf{x}_\delta$ [29]. However, it cannot be applied to our case, because $\mathbf{x}_m$ and $\mathbf{x}_\delta$ are different due to $\mathbf{F}_e$. Thus, we developed a new modification, which removes the computation of $\mathbf{x}_m$, and at the same time, is applicable to impedance control. Substituting (48) into (47) yields

$$\begin{aligned}\mathbf{v}_{\text{new}(t)} = \ddot{\mathbf{x}}_{d(t)} + \mathbf{M}_d^{-1}\left(\mathbf{B}_d\dot{\mathbf{x}}_{d(t)} + \mathbf{K}_d\mathbf{x}_{d(t)} - \mathbf{F}_{e(t)}\right) \\ - \ddot{\mathbf{x}}_{(t-L)} - \mathbf{M}_d^{-1}\left(\mathbf{B}_d\dot{\mathbf{x}}_{(t-L)} + \mathbf{K}_d\mathbf{x}_{(t-L)}\right) \\ + \left[\ddot{\mathbf{x}}_{m(t-L)} + \mathbf{M}_d^{-1}\left(\mathbf{B}_d\dot{\mathbf{x}}_{m(t-L)} + \mathbf{K}_d\mathbf{x}_{m(t-L)}\right)\right].\end{aligned} \tag{58}$$

The last three terms in the bracket are identical to the delayed value of the left-hand side terms in (57). Thus, we can replace (47) and (48) with the following modified relations:

$$\begin{aligned}\mathbf{v}_{\mathrm{new}(t)} &= \ddot{\mathbf{x}}_{d(t)} + \mathbf{M}_d^{-1}\left(\mathbf{B}_d\dot{\mathbf{x}}_{d(t)} + \mathbf{K}_d\mathbf{x}_{d(t)} - \mathbf{F}_{e(t)}\right)\\ &\quad - \left[\ddot{\mathbf{x}}_{(t-L)} + \mathbf{M}_d^{-1}\left(\mathbf{B}_d\dot{\mathbf{x}}_{(t-L)} + \mathbf{K}_d\mathbf{x}_{(t-L)}\right)\right]\\ &\quad + \mathbf{v}_{\mathrm{new}(t-L)}\end{aligned} \tag{59}$$

with

$$\mathbf{v}_{\mathrm{new}(0)} = \ddot{\mathbf{x}}_{d(0)} + \mathbf{M}_d^{-1}\left(\mathbf{B}_d\dot{\mathbf{x}}_{d(0)} + \mathbf{K}_d\mathbf{x}_{d(0)}\right) - \mathbf{F}_{e(0)}. \tag{60}$$

Therefore, $\mathbf{v}_{\mathrm{new}}$ no longer requires $\mathbf{x}_m$. Consequently, the proposed control does not need $\mathbf{x}_m$.

2) Attenuation of Sensor Resolution Effect: Position and force sensors have their own resolution [30]. Thus, the measured position and force have quantization errors due to sensor resolution as follows:

$$\begin{aligned}\hat{\mathbf{x}} &= \mathbf{x} + \mathbf{x}_\Delta\\ \hat{\mathbf{F}}_e &= \mathbf{F}_e + \mathbf{F}_{e\Delta}\end{aligned} \tag{61}$$

where $\hat{\bullet}$ denotes the measured value of $\bullet$ and $\bullet_\Delta$ denotes the quantization error. Magnitude of the quantization error lies between zero and half of the resolution [30]. The velocity $\hat{\dot{\mathbf{x}}}$ and acceleration $\hat{\ddot{\mathbf{x}}}$[2] also have quantization error inherited from that of the position sensor

$$\begin{aligned}\hat{\dot{\mathbf{x}}}_{(t)} &= \dot{\mathbf{x}}_{n(t)} + \dot{\mathbf{x}}_{\Delta(t)}\\ \hat{\ddot{\mathbf{x}}}_{(t)} &= \ddot{\mathbf{x}}_{n(t)} + \ddot{\mathbf{x}}_{\Delta(t)}\end{aligned} \tag{62}$$

where $\ddot{\mathbf{x}}_n \triangleq (\mathbf{x}_{(t)} - 2\mathbf{x}_{(t-L)} + \mathbf{x}_{(t-2L)})L^{-2}$, $\dot{\mathbf{x}}_n \triangleq (\mathbf{x}_{(t)} - \mathbf{x}_{(t-L)})L^{-1}$, $\dot{\mathbf{x}}_\Delta \triangleq (\mathbf{x}_{\Delta(t)} - \mathbf{x}_{\Delta(t-L)})L^{-1}$, and $\ddot{\mathbf{x}}_\Delta \triangleq (\mathbf{x}_{\Delta(t)} - 2\mathbf{x}_{\Delta(t-L)} + \mathbf{x}_{\Delta(t-2L)})L^{-2}$. $\dot{\mathbf{x}}_n$ and $\ddot{\mathbf{x}}_n$ become more accurate as L decreases. Conversely, $\dot{\mathbf{x}}_\Delta, \ddot{\mathbf{x}}_\Delta$ become large as L decreases. Thus, we can assume that $\dot{\mathbf{x}}_n \cong \dot{\mathbf{x}}$, $\ddot{\mathbf{x}}_n \cong \ddot{\mathbf{x}}$. Note that we consider only small time delay L, which is identical to the sampling time (order of milliseconds). In the discrete-time domain, with consideration of the quantization error, (59) can be written as follows:

$$\begin{aligned}\mathbf{v}_{\mathrm{new}(k)} &= \ddot{\mathbf{x}}_{d(k)} + \mathbf{M}_d^{-1}\left(\mathbf{B}_d\dot{\mathbf{x}}_{d(k)} + \mathbf{K}_d\mathbf{x}_{d(k)} - \mathbf{F}_{e(k)}\right)\\ &\quad + \sum_{i=0}^{k-1}\left\{\left(\ddot{\mathbf{x}}_{d(i)} - \ddot{\mathbf{x}}_{(i)}\right) + \mathbf{M}_d^{-1}\left[\mathbf{B}_d\left(\dot{\mathbf{x}}_{d(i)} - \dot{\mathbf{x}}_{(i)}\right)\right.\right.\\ &\quad \left.\left. + \mathbf{K}_d\left(\mathbf{x}_{d(i)} - \mathbf{x}_{(i)}\right) - \mathbf{F}_{e(i)}\right]\right\}\\ &\quad - \sum_{i=0}^{k-1}\left[\ddot{\mathbf{x}}_{\Delta(i)} + \mathbf{M}_d^{-1}\left(\mathbf{B}_d\dot{\mathbf{x}}_{\Delta(i)} + \mathbf{K}_d\mathbf{x}_{\Delta(i)} + \mathbf{F}_{e\Delta(i)}\right)\right].\end{aligned} \tag{63}$$

The last line of (63) represents the accumulated quantization error. By comparing (63) with (59), it can be seen that the additional IMC feedback accumulates quantization error. It implies that the accumulated error hinders precise compensation of the TDE error.

To attenuate the accumulation of quantization error, (59) is modified as

$$\begin{aligned}\mathbf{v}_{\mathrm{new}(t)} &= \ddot{\mathbf{x}}_{d(t)} + \mathbf{M}_d^{-1}(\mathbf{B}_d\dot{\mathbf{x}}_{d(t)} + \mathbf{K}_d\mathbf{x}_{d(t)} - \mathbf{F}_{e(t)})\\ &\quad + \boldsymbol{\Gamma}\{-[\ddot{\mathbf{x}}_{(t-L)} + \mathbf{M}_d^{-1}(\mathbf{B}_d\dot{\mathbf{x}}_{(t-L)} + \mathbf{K}_d\mathbf{x}_{(t-L)})]\\ &\quad + \mathbf{v}_{\mathrm{new}(t-L)}\}\end{aligned} \tag{64}$$

where $\boldsymbol{\Gamma} \in \mathfrak{R}^{\mathrm{n}\times\mathrm{n}}$ denotes a diagonal forgetting factor [31] matrix whose diagonal elements $\gamma_{ii}(i = 1, 2, \ldots, n)$ lie between zero and one: $0 < \gamma_{ii} < 1$. Similar to (63), (64) can be rewritten as

$$\begin{aligned}\mathbf{v}_{\mathrm{new}(k)} &= \ddot{\mathbf{x}}_{d(k)} + \mathbf{M}_d^{-1}\left(\mathbf{B}_d\dot{\mathbf{x}}_{d(k)} + \mathbf{K}_d\mathbf{x}_{d(k)} - \mathbf{F}_{e(k)}\right)\\ &\quad + \sum_{i=0}^{k-1}\boldsymbol{\Gamma}^{k-i}\left\{\left(\ddot{\mathbf{x}}_{d(i)} - \ddot{\mathbf{x}}_{(i)}\right) + \mathbf{M}_d^{-1}\left[\mathbf{B}_d\left(\dot{\mathbf{x}}_{d(i)} - \dot{\mathbf{x}}_{(i)}\right)\right.\right.\\ &\quad \left.\left. + \mathbf{K}_d\left(\mathbf{x}_{d(i)} - \mathbf{x}_{(i)}\right) - \mathbf{F}_{e(i)}\right]\right\} - \sum_{i=0}^{k-1}\boldsymbol{\Gamma}^{k-i}\left[\ddot{\mathbf{x}}_{\Delta(i)}\right.\\ &\quad \left. + \mathbf{M}_d^{-1}\left(\mathbf{B}_d\dot{\mathbf{x}}_{\Delta(i)} + \mathbf{K}_d\mathbf{x}_{\Delta(i)} + \mathbf{F}_{e\Delta(i)}\right)\right].\end{aligned} \tag{65}$$

It is ensured that old data, which are not representative of the current system, are discarded by exponentially decreasing weight $\boldsymbol{\Gamma}^{k-i}$ [31]. Thus, the accumulation of quantization error is attenuated. Robustness may be degraded as $\boldsymbol{\Gamma}$ reduces the IMC feedback terms in (64); thus, careful tuning of $\boldsymbol{\Gamma}$ is required.

3) Use of a Low-Pass Filter: Generally, TDE-based controls use a first-order digital low-pass filter $G_f(z)$ in the TDE loop for discrete-time domain implementation [17], [21], [32]. Thus, with $G_f(z)$, the output of TDE $\hat{\mathbf{H}}$ becomes

$$\hat{\mathbf{H}}_{(k)} = G_f(z)\mathbf{H}_{(k-1)} = G_f(z)(\mathbf{F}_{u(k-1)} - \bar{\mathbf{M}}_x\ddot{\mathbf{x}}_{(k-1)}). \tag{66}$$

Moreover, it was proved in [32] that the use of a low-pass filter has the same effect as lowering $\bar{\mathbf{M}}_x$.

IV. Experimental Studies

A. Experimental Setup

The robot used in the experiment is a 2-DOF SCARA-type robot, as shown in Fig. 5(a). The lengths of the two links are $l_1 = 0.35$ m and $l_2 = 0.294$ m, respectively; their masses are $m_1 = 11.17$ kg and $m_2 = 6.82$ kg; the distance from the joint axis to the center of mass for each link is $L_1 = 0.30$ m and $L_2 = 0.18$ m, respectively; the moment of inertia about the joint axis is $I_1 = 1.0\ \mathrm{kg\cdot m^2}$ and $I_2 = 0.224\ \mathrm{kg\cdot m^2}$, respectively. At joint 1, a brushless dc (BLDC) motor with a stall torque of 2.39 N·m is used to transmit power through a harmonic drive with a gear reduction ratio of 100:1, while at joint 2, a BLDC motor having a stall torque of 0.92 N·m is used with a gear reduction ratio of 80:1. Each joint has a resolver attached at its shaft for sensing the angular displacement with a resolution of 4096 pulses/revolution. An ATI Gamma SI-130-10 force sensor, having 130 N range and 0.1 N resolution, is attached at the end-effector. For the environment, an Al plate with a stiffness of

[2] Backward Euler differentiation is used.

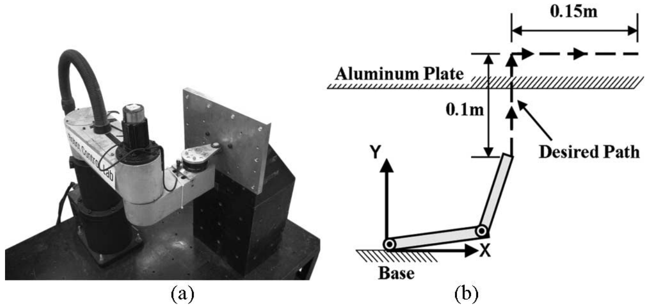

Fig. 5. SCARA-type robot under impedance control. (a) SCARA-type robot system. (b) Schematic diagram of the experimental setup.

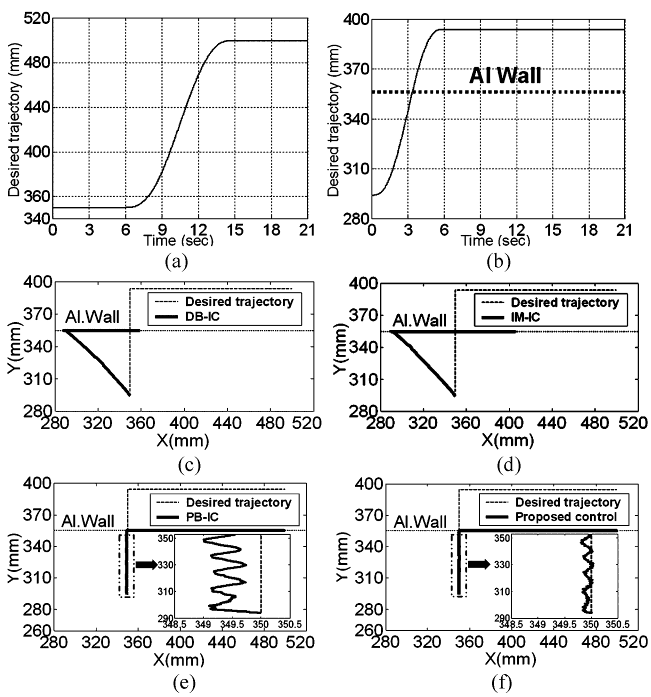

Fig. 6. Position response of each control without a payload. (a) x-axis desired trajectory. (b) y-axis desired trajectory. (c) DB-IC. (d) IM-IC. (e) PB-IC. (f) Proposed control.

about 150 000 N/m is used. The implementation of the controller was made in QNX, a real-time operating system environment, with a sampling frequency of 1 kHz.

B. Task

To demonstrate the performance of different control methods, we considered a simple task that involves contact with the Al plate. The trajectory, illustrated in Figs. 5(b) and 6(a) and 6(b), is designed as follows: from 0 to 6 s, a y-axis 0.1-m straight line trajectory, in the form of a fifth-order polynomial, toward the Al plate is given; then from 6 to 15 s, an x-axis 0.15-m straight line trajectory, in the form of a fifth-order polynomial, is given; finally, from 15 to 21 s, the end point of the second trajectory is designated with zero velocity and acceleration.

In order to verify the robustness against modeling error, the same task is performed with two conditions: 1) without a payload and 2) with a 3.11-kg payload, corresponding to nearly half of the second link mass, at the end-effector side.

TABLE I
RESPONSE OF EACH CONTROL WITHOUT A PAYLOAD

Control	DB-IC	IM-IC	PB-IC	Proposed control
Max.deviation (*mm*)	61.95	60.72	1.03	0.21
Amplitude of force fluctuation (*N*)	22.0	12.6	32.1	9.8

C. Desired Impedance and Gains

Desired impedance parameters are designed as follows: $\mathbf{M}_d = \mathrm{diag}(20, 20)$kg, $\mathbf{B}_d = \mathrm{diag}(900, 900)$N·s/m, $\mathbf{K}_d = \mathrm{diag}(400, 400)$N/m. An overdamped impedance dynamics, a damping ratio of 5, is selected. Besides the target dynamics, the following gains are used for each controller:

1) for PB-IC: $\mathbf{K}_v = \mathrm{diag}(90, 90)$, $\mathbf{K}_p = \mathrm{diag}(2500, 2500)$;
2) for IM-IC: $\mathbf{K}_v = \mathrm{diag}(150, 150)$, $\mathbf{K}_p = \mathrm{diag}(5625, 5625)$;
3) for the proposed control: $\bar{\mathbf{M}}_x = \mathbf{J}^{-T}\bar{\mathbf{M}}_\theta\mathbf{J}^{-1}$, $\bar{\mathbf{M}}_\theta = \mathrm{diag}(0.03, 0.0051)$, $\mathbf{\Gamma} = \mathrm{diag}(0.9, 0.83)$.

For PB-IC and IM-IC, these gains were tuned to minimize the inner loop tracking error through extensive experiments.

The form of $\bar{\mathbf{M}}_x$ is obtained from the relation: $\mathbf{M}_x = \mathbf{J}^{-T}\mathbf{M}_\theta\mathbf{J}^{-1}$ [22]. The $\bar{\mathbf{M}}_\theta$, known part of $\mathbf{M}_\theta$, is a diagonal constant matrix. Note that the Jacobian, the kinematic relation, is easy to obtain, compared with the difficulty of obtaining dynamics, i.e., Coriolis force, centrifugal force, gravity, and friction force. Different values of the forgetting factor are selected for each axis. Only the y-axis motion is constrained by the Al plate. Thus, large attenuation (small forgetting factor) of the y-axis quantization error was required for smooth execution of the task. In contrast, the x-axis motion is free motion; thus, small attenuation (a large forgetting factor) was sufficient.

D. Experimental Results

1) Without Payload: The x–y position response is shown in Fig. 6. In free space, the proposed controller shows the smallest position deviation, whereas DB-IC presents the largest (see Table I). Note that PB-IC shows remarkably improved position tracking performance compared with that of DB-IC.

Strictly speaking, the task cannot be executed with DB-IC or IM-IC. The position deviation in free space is too large. Thus, DB-IC and IM-IC are excluded from further comparison.

The y-axis force response is shown in Fig. 7. The proposed control shows a small impact force (57.6 N) compared with that (70.2 N) of PB-IC. The proposed control also shows the smallest amplitude of force fluctuation (see Table I) while moving along a section of the plate from 6 to 15 s. In contrast, the robot under PB-IC loses contact with wall six times, and shows large force fluctuation.

To verify the accuracy, the impedance error $\boldsymbol{\sigma}_{(t)} \in \mathfrak{R}^{\mathrm{n}}$, defined in (67), is considered. The velocity and acceleration are calculated from a sophisticated offline numerical differentiation

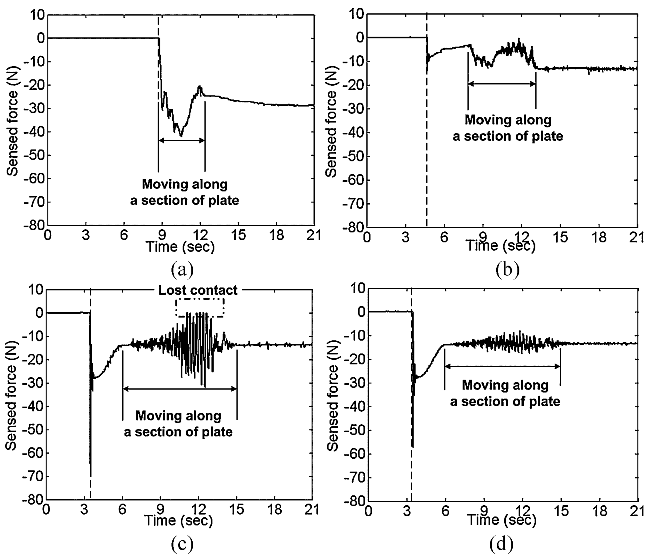

Fig. 7. y-axis force response of each control without a payload. Dashed line indicates the contact time. (a) DB-IC. (b) IM-IC. (c) PB-IC. (d) Proposed control.

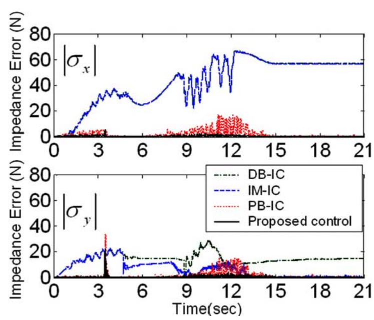

Fig. 8. Impedance error of each control without a payload.

[33] of position response.

$$\boldsymbol{\sigma}_{(t)} \triangleq \mathbf{M}_d(\ddot{\mathbf{x}}_{d(t)} - \ddot{\mathbf{x}}_{(t)}) + \mathbf{B}_d(\dot{\mathbf{x}}_{d(t)} - \dot{\mathbf{x}}_{(t)}) + \mathbf{K}_d(\mathbf{x}_{d(t)} - \mathbf{x}_{(t)}) - \mathbf{F}_{e(t)}. \quad (67)$$

If $\boldsymbol{\sigma}_{(t)} = [\sigma_x \quad \sigma_y]^T$ is a zero vector, then (67) becomes the desired impedance dynamics in (1). The x- and y-axis absolute values of impedance error, $|\sigma_x|$ and $|\sigma_y|$, are shown in Fig. 8. It is clear that the proposed control yields the smallest impedance error throughout the task. Therefore, the proposed control can accurately realize desired impedance.

At this point, it appears relevant to make a final analysis on the difference of impedance error, position error, and force fluctuation due to each control method mentioned earlier. It is easy to derive the relationships as follows:

$$\boldsymbol{\sigma} = \mathbf{M}_d\boldsymbol{\eta} \qquad \text{for DB-IC} \quad (68)$$

$$\begin{aligned}\boldsymbol{\sigma} &= \mathbf{Z}_d(s)\mathbf{C}^{-1}(s)\boldsymbol{\eta} \\ &= \mathbf{M}_d[s^2\mathbf{I} + \mathbf{M}_d^{-1}(s\mathbf{B}_d + \mathbf{K}_d)]\mathbf{C}^{-1}(s)\boldsymbol{\eta} \\ &\quad \text{for PB-IC}\end{aligned} \quad (69)$$

$$\boldsymbol{\sigma} = \mathbf{M}_d(1 - e^{-Ls})\boldsymbol{\eta}_T \qquad \text{for proposed control.} \quad (70)$$

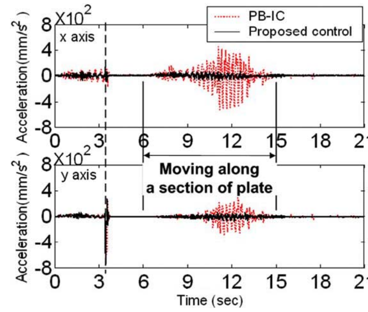

Fig. 9. Acceleration responses without a payload. Dashed line indicates the contact time. x-axis acceleration causes y-axis acceleration due to the dynamic coupling effect.

These relationships clearly show that how each $\boldsymbol{\sigma}$ is determined for given dynamics estimation error $\boldsymbol{\eta}$ (or $\boldsymbol{\eta}_T$). Of course, (54)–(56) show in essence the same fact, yet these are more immediate and direct. The proposed control could achieve the smallest impedance error due to $(1 - e^{-Ls})$, which greatly reduces the effect of $\boldsymbol{\eta}$, whereas DB-IC showed the largest error because $\boldsymbol{\eta}$ is transmitted to $\boldsymbol{\sigma}$ with no such an attenuation mechanism. Note that PB-IC has some attenuation due to $\mathbf{Z}_d(s)\mathbf{C}^{-1}(s)$, but it is not so effective as $(1 - e^{-Ls})$. In the case of the proposed control, if the feedback time delay L goes to zero, $\boldsymbol{\eta}$ will be perfectly removed.

In addition, the aforementioned relationships also explain why there are differences in free space position errors due to the three control methods. One can express the free-space closed-loop error dynamics due to each control method as follows:

$$\mathbf{Z}_d(s)(\mathbf{x}_d - \mathbf{x}) = \mathbf{M}_d\boldsymbol{\eta} \qquad \text{for DB-IC} \quad (71)$$

$$\begin{aligned}\mathbf{Z}_d(s)(\mathbf{x}_d - \mathbf{x}) &= \mathbf{Z}_d(s)\mathbf{C}^{-1}(s)\boldsymbol{\eta} \\ &= \mathbf{M}_d[s^2\mathbf{I} + \mathbf{M}_d^{-1}(s\mathbf{B}_d + \mathbf{K}_d)]\mathbf{C}^{-1}(s)\boldsymbol{\eta} \\ &\quad \text{for PB-IC}\end{aligned} \quad (72)$$

$$\mathbf{Z}_d(s)(\mathbf{x}_d - \mathbf{x}) = \mathbf{M}_d(1 - e^{-Ls})\boldsymbol{\eta}_T \quad \text{for proposed control.} \quad (73)$$

Therefore, of the three, the tracking error due to the proposed method is the smallest and that due to DB-IC the largest.

Further, the force fluctuation while moving along a section of plate (from 6 to 15 s) can also be explained with aforementioned analysis. The fluctuation comes from *dynamic coupling* [21], [22], [26]: coupling of dynamics of two axes. Specifically, from (67)–(70), $\boldsymbol{\eta}$ couples motions of the two axes. An examination of acceleration responses in Fig. 9 confirms that, from 6 to 15 s, the fluctuation of x-axis acceleration induces y-axis acceleration. It implies that y-axis force fluctuation is induced by x-axis acceleration through *dynamic coupling*. Thus, the fluctuation strongly relies on the compensation performance of $\boldsymbol{\eta}$.

In addition, in the case of PB-IC, from (69), $\boldsymbol{\eta}$ excites inner loop dynamics $\mathbf{C}^{-1}(s)$, which is different from desired impedance dynamics. Therefore, amplitude of force fluctuation (and amplitude of acceleration fluctuation) due to the proposed control is the smallest and that due to PB-IC the largest. In the case of the proposed control, it was found that the small

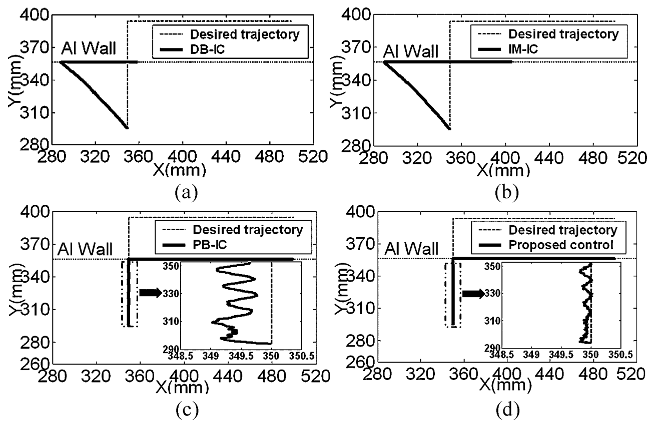

Fig. 10. Position response of each control with a 3.11-kg payload. Dotted line indicates the wall. (a) DB-IC. (b) IM-IC. (c) PB-IC. (d) Proposed control.

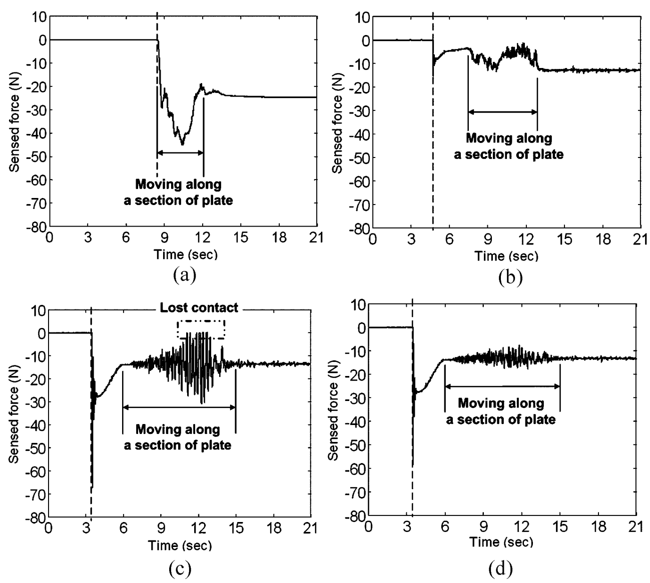

Fig. 11. Force response of each control with a 3.11-kg payload. Dashed line indicates the contact time. (a) DB-IC. (b) IM-IC. (c) PB-IC. (d) Proposed control.

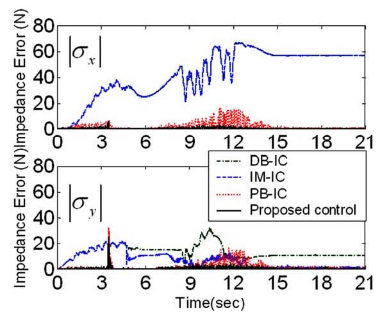

Fig. 12. Impedance error of each control with a 3.11-kg payload.

force fluctuation is dominated by the torque ripple of joint motor/driver [21]. This will be discussed in Section V-B.

2) With Payload: We performed the same task with an additional 3.11-kg payload at the end-effector side. The x–y position response, y-axis force response, and impedance error are shown in Figs. 10–12, respectively. In this case as well, position deviation, force fluctuation, and impedance error due to the proposed control are the smallest, thereby demonstrating its accuracy and robustness against the modeling error.

V. Comparison With the Disturbance Observer

A. Comparison of Structure

The DOB evaluates the effect of all the uncertainties, the estimate of which is denoted by $\mathbf{F}_{\rm dis} \in \Re^{\rm n}$, as follows:

$$\mathbf{F}_{\rm dis} = \mathbf{F}_u - \mathbf{M}_{nn}\ddot{\mathbf{x}} \tag{74}$$

where $\mathbf{M}_{nn} \in \Re^{\rm n\times n}$ denotes the Cartesian space standard mass matrix [23]. In the continuous-time domain, $\mathbf{F}_{\rm dis}$, after processed with a first-order low-pass filter $G_H(s)$, becomes $\hat{\mathbf{F}}_{\rm dis}$, the output of DOB, as follows:

$$\hat{\mathbf{F}}_{\rm dis} = G_H(s)(\mathbf{F}_u - \mathbf{M}_{nn}\ddot{\mathbf{x}}). \tag{75}$$

The impedance control with DOB has the following form [23]:

$$\mathbf{F}_u = \mathbf{M}_{nn}(\mathbf{M}_{vn}^{-1}\mathbf{F}^{\rm ref}) + \hat{\mathbf{F}}_{\rm dis} \tag{76}$$

where $\mathbf{M}_{vn} = \mathbf{M}_d$ and $\mathbf{F}^{\rm ref} = \mathbf{M}_d\ddot{\mathbf{x}}_d + \mathbf{B}_d(\dot{\mathbf{x}}_d - \dot{\mathbf{x}}) + \mathbf{K}_d(\mathbf{x}_d - \mathbf{x}) - \mathbf{F}_e$. Since DOB was implemented in the discrete-time domain [23]–[25], the control law in fact is

$$\mathbf{F}_{u(k)} = \mathbf{M}_{nn}(\mathbf{M}_{vn}^{-1}\mathbf{F}_{(k)}^{\rm ref}) + \underbrace{G_{HZ}(z)\left(\mathbf{F}_{u(k-1)} - \mathbf{M}_{nn}\ddot{\mathbf{x}}_{(k-1)}\right)}_{\text{DOB output } \hat{\rm F}_{\rm dis}} \tag{77}$$

where $G_{HZ}(z)$ denotes the first-order digital low-pass filter. Thus, from (66) and (77), one can immediately confirm that the structure of DOB and that of TDE is equivalent. However, the impedance control with DOB has no mechanism to reduce the effect of uncompensated uncertainties $(\mathbf{F}_{\rm dis} - \hat{\mathbf{F}}_{\rm dis})$. In contrast, the proposed control has IMC feedback loop, which corrects the effect of uncompensated dynamics estimation error $\boldsymbol{\eta}_T$ ($\boldsymbol{\eta}_T = \bar{\mathbf{M}}_x^{-1}(\mathbf{H} - \hat{\mathbf{H}})$).

B. Experimental Comparison of Performance

The experimental setup, trajectory, and desired impedance parameters are the same as those in Section IV. A payload was not attached to the robot. The gains of the proposed control are the same as those tuned in Section IV.

The gains of the impedance control with DOB were tuned by following the guideline in [23]–[25]. Originally, the form of $\mathbf{M}_{nn}$ was a diagonal matrix, which is defined at the initial configuration of the robot [23]. However, it was reported in [23] that with a diagonal $\mathbf{M}_{nn}$, closed-loop dynamics is sensitive to the variation of $\mathbf{M}_x$, and if the variation is large, then the closed-loop dynamics becomes unstable [23]. This experiment belongs to such a case: the final value of each element of $\mathbf{M}_x$ is more than two times larger than the initial values in $\mathbf{M}_x$. Thus, a more accurate $\mathbf{M}_{nn}$, $\mathbf{J}^{-T}[\mathrm{diag}(I_1, I_2)]\mathbf{J}^{-1}$, was used. I_1 and I_2, given in Section IV. A, denote the nominal inertia of each joint. Note that this form is the same as the form of $\bar{\mathbf{M}}_x$.

The cutoff frequency of $G_{HZ}(z)$ was then set to be 100 rad/s. When the cutoff frequency was increased over 100 rad/s, there

TABLE II
SUMMARY OF RESPONSE OF EACH CONTROL

	Impedance control with DOB ①	Proposed control ②	②/① ×100(%)
Max. deviation (*mm*)	0.36	0.17	47.2%
Amplitude of force fluctuation (*N*)	10.01	7.59	75.8%

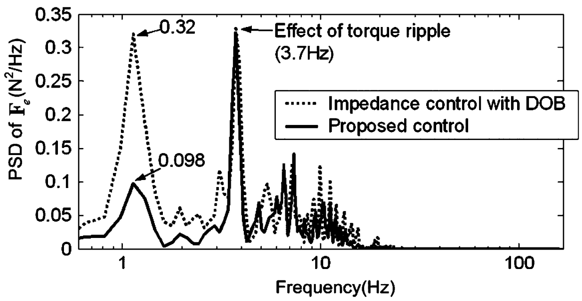

Fig. 13. PSD of $\mathbf{F}_e$ due to each control during the constrained motion (from 6 to 15 s). Note that, before computing PSD, the mean value (dc part) of $\mathbf{F}_e$ due to each control is subtracted from $\mathbf{F}_e$. Due to the torque ripple of the BLDC motor/driver system, at 3.7 Hz, the largest peak occurs. Below 3.7 Hz, PSD of $\mathbf{F}_e$ due to the proposed control is about 30% of that due to the impedance control with DOB.

were no further improvements of performance; instead, loud noise accompanied at the robot joints.

From the responses summarized in Table II, one can see that, due to the IMC loop, the performance of the proposed control is superior to that of the impedance control with DOB. However, the relatively small difference between amplitudes of force fluctuation due to the two controllers may call for some explanations. A careful examination of the power spectral density (PSD) of $\mathbf{F}_e$ (see Fig. 13) during the constrained motion (from 6 to 15 s) reveals that the frequency of *torque ripple* of the joint BLDC motor/driver system [21] and the frequency of the largest peak are almost the same (about 3.7 Hz). Thus, it can be concluded that force fluctuation due to the two controllers is dominated by the effect of torque ripple. Note that, below 3.7 Hz, the PSD of $\mathbf{F}_e$ of the proposed control is about 30% of that of the impedance control with DOB.

VI. Conclusion

In this paper, as a solution to the "accuracy/robustness dilemma in impedance control," both accurate and robust impedance control using an IMC structure and TDE was proposed. The suggested approach has a simple structure, which is straightforward to design and easy to implement without entire robot model computation or complex algorithms. The proposed control is accurate and robust owing to dynamics estimation error correction using the IMC structure and efficient estimation of nonlinear dynamics using TDE.

The accuracy and robustness of the proposed controller are verified with a 2-DOF SCARA-type robot and a stiff Al wall. Even with a 3.11-kg payload, the proposed controller shows accurate and robust performance throughout the task.

In addition, the difference between the proposed control and the impedance control with DOB is clarified and experimental comparisons are made. It turns out that the proposed control has an additional IMC feedback loop, which makes improvement to this extent over that of the impedance control with DOB.

Therefore, we believe that the proposed control is a simple yet effective solution to the "accuracy/robustness dilemma in impedance control."

Appendix

Here, the closed-loop stability will be analyzed when the proposed control is used. Meaning of stability condition related to the convergence is then discussed.

The analysis is similar to the proof of [21]. The control input (52) becomes

$$\mathbf{u}_{\mathrm{new}(t)} = \mathbf{u}_{d(t)} + \boldsymbol{\Gamma}\boldsymbol{\eta}_{T(t-L)}. \tag{78}$$

By substituting (78) into (42), the closed-loop dynamics due to the proposed control becomes

$$\begin{aligned}(\ddot{\mathbf{x}}_{d(t)} - \ddot{\mathbf{x}}_{(t)}) + \mathbf{M}_d^{-1}[\mathbf{B}_d(\dot{\mathbf{x}}_{d(t)} - \dot{\mathbf{x}}_{(t)})\\ +\mathbf{K}_d(\mathbf{x}_{d(t)} - \mathbf{x}_{(t)}) - \mathbf{F}_{e(t)}] = (\boldsymbol{\eta}_{T(t)} - \boldsymbol{\Gamma}\boldsymbol{\eta}_{T(t-L)}).\end{aligned} \tag{79}$$

If $\boldsymbol{\eta}_{T(t)}$ is bounded, then the closed-loop dynamics (79) is bounded, and consequently, the overall system is stable. Therefore, the boundedness of $\boldsymbol{\eta}_{T(t)}$ is addressed in detail.

Stability Theorem: Consider an n-DOF robot under the proposed control. Assume that the desired trajectory, its time derivatives, and the external force are in L_∞^n space ($\mathbf{x}_d, \dot{\mathbf{x}}_d, \ddot{\mathbf{x}}_d \in L_\infty^n$, and $\mathbf{F}_e \in L_\infty^n$). The closed-loop system under the proposed control is then L_∞^n-stable when the following stability condition is satisfied.

Stability condition

$$\left\| \left\| \mathbf{I} - \mathbf{M}_{x(t)}^{-1}\bar{\mathbf{M}}_x \right\|_{i2} \right\|_\infty < (\|\boldsymbol{\Gamma} + \mathbf{I}\|_{i2} + \|\boldsymbol{\Gamma}\|_{i2})^{-1} \tag{80}$$

where $\|\bullet\|_{i2}$ denotes induced 2-norm of a matrix $\bullet$ and $\|\bullet\|_\infty$ denotes L_∞^n-norm of $\bullet_{(t)}$ [34].

Proof of Theorem: Equation (79) can be written as

$$(\boldsymbol{\eta}_{T(t)} - \boldsymbol{\Gamma}\boldsymbol{\eta}_{T(t-L)}) = \mathbf{u}_{d(t)} - \ddot{\mathbf{x}}_{(t)}. \tag{81}$$

Multiplying $\mathbf{M}_{x(t)}$ on both sides of (81) and substituting (5) into it leads to

$$\begin{aligned}\mathbf{M}_{x(t)}(\boldsymbol{\eta}_{T(t)} - \boldsymbol{\Gamma}\boldsymbol{\eta}_{T(t-L)}) = \mathbf{M}_{x(t)}\mathbf{u}_{d(t)}\\ +(\mathbf{N}_{x(t)} + \mathbf{F}_{e(t)} - \mathbf{F}_{u(t)}).\end{aligned} \tag{82}$$

Meanwhile, from (31) and (5), we obtain

$$\begin{aligned}\mathbf{H}_{(t-L)} &= -\bar{\mathbf{M}}_x\ddot{\mathbf{x}}_{(t-L)} + \mathbf{F}_{u(t-L)}\\ &= (\mathbf{M}_{x(t-L)} - \bar{\mathbf{M}}_x)\ddot{\mathbf{x}}_{(t-L)} + \mathbf{N}_{x(t-L)} + \mathbf{F}_{e(t-L)}.\end{aligned} \tag{83}$$

Substitution of (78) and (83) into (45) yields

$$\begin{aligned}\mathbf{F}_{u(t)} &= \bar{\mathbf{M}}_x\mathbf{u}_{d(t)} + \bar{\mathbf{M}}_x\boldsymbol{\Gamma}\boldsymbol{\eta}_{T(t-L)}\\ &\quad + (\mathbf{M}_{x(t-L)} - \bar{\mathbf{M}}_x)\ddot{\mathbf{x}}_{(t-L)} + \mathbf{N}_{x(t-L)} + \mathbf{F}_{e(t-L)}.\end{aligned} \tag{84}$$

Eliminating $\mathbf{F}_{u(t)}$ in (82) and (84) leads to

$$\mathbf{M}_{x(t)}(\boldsymbol{\eta}_{T(t)} - \boldsymbol{\Gamma}\boldsymbol{\eta}_{T(t-L)}) = (\mathbf{M}_{x(t)} - \bar{\mathbf{M}}_x)\mathbf{u}_{d(t)} - \bar{\mathbf{M}}_x\boldsymbol{\Gamma}\boldsymbol{\eta}_{T(t-L)} - (\mathbf{M}_{x(t-L)} - \bar{\mathbf{M}}_x)\ddot{\mathbf{x}}_{(t-L)} + \mathbf{N}_{x(t)} - \mathbf{N}_{x(t-L)} + \mathbf{F}_{e(t)} - \mathbf{F}_{e(t-L)}. \quad (85)$$

Since $\ddot{\mathbf{x}}_{(t-L)} = \mathbf{u}_{d(t-L)} - (\boldsymbol{\eta}_{T(t-L)} - \boldsymbol{\Gamma}\boldsymbol{\eta}_{T(t-2L)})$ from (81), (85) becomes

$$\begin{aligned}\mathbf{M}_{x(t)}&(\boldsymbol{\eta}_{T(t)} - \boldsymbol{\Gamma}\boldsymbol{\eta}_{T(t-L)})\\ &= (\mathbf{M}_{x(t)} - \bar{\mathbf{M}}_x)(\boldsymbol{\eta}_{T(t-L)} - \boldsymbol{\Gamma}\boldsymbol{\eta}_{T(t-2L)}) - \bar{\mathbf{M}}_x\boldsymbol{\Gamma}\boldsymbol{\eta}_{T(t-L)}\\ &\quad + (\mathbf{M}_{x(t)} - \bar{\mathbf{M}}_x)(\mathbf{u}_{d(t)} - \mathbf{u}_{d(t-L)}) + (\mathbf{M}_{x(t)} - \mathbf{M}_{x(t-L)})\\ &\quad \times \ddot{\mathbf{x}}_{(t-L)} + \mathbf{N}_{x(t)} - \mathbf{N}_{x(t-L)} + \mathbf{F}_{e(t)} - \mathbf{F}_{e(t-L)}. \qquad (86)\end{aligned}$$

From (86), the TDE error $\boldsymbol{\eta}_{T_{(t)}}$ becomes

$$\boldsymbol{\eta}_{T(t)} = (\mathbf{I} - \mathbf{M}_{x(t)}^{-1}\bar{\mathbf{M}}_x)[(\boldsymbol{\Gamma} + \mathbf{I})\boldsymbol{\eta}_{T(t-L)} - \boldsymbol{\Gamma}\boldsymbol{\eta}_{T(t-2L)}] + \boldsymbol{\xi}_{1(t-L)} + (\mathbf{I} - \mathbf{M}_{x(t)}^{-1}\bar{\mathbf{M}}_x)\,\boldsymbol{\xi}_{2(t-L)} \quad (87)$$

where

$$\boldsymbol{\xi}_{1(t-L)} = \mathbf{M}_{x(t)}^{-1}[(\mathbf{M}_{x(t)} - \mathbf{M}_{x(t-L)})\ddot{\mathbf{x}}_{(t-L)} + \mathbf{N}_{x(t)} - \mathbf{N}_{x(t-L)} + \mathbf{F}_{e(t)} - \mathbf{F}_{e(t-L)}] \quad (88)$$

$$\boldsymbol{\xi}_{2(t-L)} = (\mathbf{u}_{d(t)} - \mathbf{u}_{d(t-L)}) \quad (89)$$

which are bounded for a sufficiently small L [21]. Taking the norms of both sides of (87), we can then derive the following inequality, where $\|\bullet\|_{T\infty}$ denotes truncated L_∞^n-norm of $\bullet_{(t)}$ [34]:

$$\|\boldsymbol{\eta}_{T(t)}\|_{T\infty} \le \mu\,\|\boldsymbol{\Gamma} + \mathbf{I}\|_{i2}\,\|\boldsymbol{\eta}_{T(t-L)}\|_{T\infty} + \mu\,\|\boldsymbol{\Gamma}\|_{i2}\,\|\boldsymbol{\eta}_{T(t-2L)}\|_{T\infty} + \|\boldsymbol{\xi}_{1(t-L)}\|_\infty + \mu\,\|\boldsymbol{\xi}_{2(t-L)}\|_\infty \quad (90)$$

where $\mu \triangleq \left\|\left\|\mathbf{I} - \mathbf{M}_{x(t)}^{-1}\bar{\mathbf{M}}_x\right\|_{i2}\right\|_\infty$.

Since $\|\bullet_{(t)}\|_{T\infty} \ge \|\bullet_{(t-L)}\|_{T\infty}$ [34], we get

$$\|\boldsymbol{\eta}_{T(t)}\|_{T\infty} \le \mu\,\|\boldsymbol{\Gamma} + \mathbf{I}\|_{i2}\,\|\boldsymbol{\eta}_{T(t)}\|_{T\infty} + \mu\,\|\boldsymbol{\Gamma}\|_{i2}\,\|\boldsymbol{\eta}_{T(t)}\|_{T\infty} + \|\boldsymbol{\xi}_{1(t-L)}\|_\infty + \mu\,\|\boldsymbol{\xi}_{2(t-L)}\|_\infty. \quad (91)$$

Rearranging (91) leads to

$$[1 - \mu\,(\|\boldsymbol{\Gamma} + \mathbf{I}\|_{i2} + \|\boldsymbol{\Gamma}\|_{i2})]\,\|\boldsymbol{\eta}_{T(t)}\|_{T\infty} \le \|\boldsymbol{\xi}_{1(t-L)}\|_\infty + \mu\,\|\boldsymbol{\xi}_{2(t-L)}\|_\infty. \quad (92)$$

The sufficient condition for this inequality to have an upper bound is shown as

$$\left\|\left\|\mathbf{I} - \mathbf{M}_{x(t)}^{-1}\bar{\mathbf{M}}_x\right\|_{i2}\right\|_\infty < (\|\boldsymbol{\Gamma} + \mathbf{I}\|_{i2} + \|\boldsymbol{\Gamma}\|_{i2})^{-1}. \quad (93)$$

If (93) holds, $\|\boldsymbol{\eta}_{T(t)}\|_\infty$ has an upper bound, because no terms of (93) are related to T, which is a symbol of a truncated norm. The fact that $\|\boldsymbol{\eta}_{T(t)}\|_\infty$ has an upper bound means that the TDE error is bounded. Therefore, the closed-loop dynamics due to the proposed control is L_∞^n-stable, if we set the control gains to satisfy (93). □

The proposed control is stable if $\bar{\mathbf{M}}_x = \mathbf{M}_x$, since it makes the left-hand side of (80) zero. In reality, however, it is difficult to estimate robot inertia $\mathbf{M}_x$ accurately, as the DOFs of a robot increases. When the $\bar{\mathbf{M}}_x$ differs from the actual inertia $\mathbf{M}_x$, the stable range of $\bar{\mathbf{M}}_x$ is determined by the forgetting factor matrix $\boldsymbol{\Gamma}$. The right-hand side of (80) is inversely proportional to $\|\boldsymbol{\Gamma}\|_{i2}$; and if the IMC feedback is not used (i.e., $\boldsymbol{\Gamma} = \mathbf{0}_n$), the stable region of $\bar{\mathbf{M}}_x$ is the largest. It is noteworthy that enhancing the robustness (use of a large forgetting factor) involves decreasing the stable region, which comes from the conflicting nature between stability and robustness in feedback control.

References

[1] N. Hogan, "Impedance control: An approach to manipulation," *Trans. ASME, J. Dyn., Syst. Meas., Control*, vol. 107, pp. 1–24, Mar. 1985.

[2] N. Hogan, "Stable execution of contact tasks using impedance control," in *Proc. IEEE ICRA*, Raleigh, NC, Mar. 1987, vol. 4, pp. 1047–1054.

[3] T. Yoshikawa, "Force control of robot manipulators," in *Proc. IEEE ICRA*, San Francisco, CA, Apr. 2000, pp. 220–226.

[4] L. Masia, H. I. Krebs, P. Cappa, and N. Hogan, "Design and characterization of hand module for whole-arm rehabilitation following stroke," *IEEE/ASME Trans. Mechatronics*, vol. 12, no. 4, pp. 399–407, Aug. 2007.

[5] M. A. Al-Mouhamed, O. Toker, and A. Iqbal, "A multithreaded distributed telerobotic framework," *IEEE/ASME Trans. Mechatronics*, vol. 11, no. 5, pp. 558–566, Oct. 2006.

[6] I. G. Polushin, P. X. Liu, and C.-H. Lung, "A force-reflection algorithm for improved transparency in bilateral teleoperation with communication delay," *IEEE/ASME Trans. Mechatronics*, vol. 12, no. 3, pp. 361–374, Jun. 2007.

[7] K. Harada, S. Kajita, F. Kanehiro, K. Fujiwara, K. Kaneko, K. Yokoi, and H. Hirukawa, "Real-time planning of humanoid robot's gait for force-controlled manipulation," *IEEE/ASME Trans. Mechatronics*, vol. 12, no. 1, pp. 53–62, Feb. 2007.

[8] L. Zollo, S. Roccella, E. Guglielmelli, M. C. Carrozza, and P. Dario, "Biomechatronic design and control of an anthropomorphic artificial hand for prosthetic and robotic applications," *IEEE/ASME Trans. Mechatronics*, vol. 12, no. 4, pp. 418–429, Aug. 2007.

[9] N. Yu, C. Hollnagel, A. Blickenstorfer, S. S. Kollias, and R. Riener, "Comparison of MRI-compatible mechatronic systems with hydrodynamic and pneumatic actuation," *IEEE/ASME Trans. Mechatronics*, vol. 13, no. 3, pp. 268–277, Jun. 2008.

[10] T. Valency and M. Zacksenhouse, "Accuracy/robustness dilemma in impedance control," *Trans. ASME, J. Dyn., Syst. Meas., Control*, vol. 125, pp. 310–319, Sep. 2003.

[11] S. Chiaverini, B. Siciliano, and L. Villani, "A survey of robot interaction control schemes with experimental comparison," *IEEE/ASME Trans. Mechatronics*, vol. 4, no. 3, pp. 273–285, Sep. 1999.

[12] D. A. Lawrence, "Impedance control stability properties in common implementations," in *Proc. IEEE ICRA*, Philadelphia, PA, Apr. 24–29, 1998, vol. 2, pp. 1185–1190.

[13] S. P. Chan and H. P. Chen, "Robust implementation of generalized impedance control for robot manipulators," *Adv. Robot.*, vol. 15, pp. 641–661, 2001.

[14] M.-C. Chien and A.-C. Huang, "Adaptive impedance control of robot manipulators based on function approximation technique," *Robotica*, vol. 22, pp. 395–403, 2004.

[15] C.-C. Cheah and D. Wang, "Learning impedance control for robot manipulators," *IEEE Trans. Robot. Autom.*, vol. 14, no. 3, pp. 452–465, Jun. 1998.

[16] C. G. Economou and M. Morari, "Internal model control. 1. A unifying review and some new results," *Ind. Eng. Chem. Process Des. Dev.*, vol. 21, no. 2, pp. 308–323, Jul. 1982.

[17] K. Youcef-Toumi and S.-T. Wu, "Input/output linearization using time delay control," *Trans. ASME, J. Dyn., Syst. Meas., Control*, vol. 114, pp. 10–19, 1992.

[18] K. H. Chan, J. Bao, and W. J. Whiten, "A new approach to control of MIMO processes with static nonlinearities using an extended IMC framework," *Comput. Chem. Eng.*, vol. 30, pp. 329–342, Dec. 2005.

[19] Q. Li, A. N. Poo, and C. M. Lim, "Internal model structure in the control of robot manipulators," *Mechatronics*, vol. 6, pp. 571–590, Aug. 1996.

[20] P. H. Chang, J. W. Lee, and S. H. Park, "Time delay observer: A robust observer for nonlinear plants," *Trans. ASME, J. Dyn., Syst. Meas., Control*, vol. 119, pp. 521–527, Sep. 1997.

[21] P. H. Chang, D. S. Kim, and K. C. Park, "Robust force/position control of a robot manipulator using time-delay control," *Control Eng. Pract.*, vol. 3, pp. 1255–1264, 1995.
[22] P. H. Chang, B. S. Park, and K. C. Park, "An experimental study on improving hybrid position/force control of a robot using time delay control," *Mechatronics*, vol. 6, pp. 915–931, 1996.
[23] T. Murakami, N. Oda, Y. Miyasaka, and K. Ohnishi, "A motion control strategy based on equivalent mass matrix in multidegree-of-freedom manipulator," *IEEE Trans. Ind. Electron.*, vol. 42, no. 2, pp. 123–130, Apr. 1995.
[24] K. Ohnishi, M. Shibata, and T. Murakami, "Motion control for advanced mechatronics," *IEEE/ASME Trans. Mechatronics*, vol. 1, no. 1, pp. 56–67, Mar. 1996.
[25] H. Kobayashi, S. Katsura, and K. Ohnishi, "An analysis of parameter variations of disturbance observer for motion control," *IEEE Trans. Ind. Electron.*, vol. 54, no. 6, pp. 3413–3421, Dec. 2007.
[26] O. Khatib, "A unified approach for motion and force control of robot manipulators: The operational space formulation," *IEEE Trans. Robot. Autom.*, vol. RA-3, no. 1, pp. 43–53, Feb. 1987.
[27] T. Umeno and Y. Hori, "Robust speed control of DC servomotors using modern two degrees-of-freedom controller design," *IEEE Trans. Ind. Electron.*, vol. 38, no. 5, pp. 363–368, Oct. 1991.
[28] R.-E. Precup, S. Preitl, I. J. Rudas, M. L. Tomescu, and J. K. Tar, "Design and experiments for a class of fuzzy controlled servo systems," *IEEE/ASME Trans. Mechatronics*, vol. 13, no. 1, pp. 22–35, Feb. 2008.
[29] M. R. Mataušek, D. M. Miljković, and B. I. Jeftenić, "Nonlinear multi-input-multi-output neural network control of DC motor drive with field weakening," *IEEE Trans. Ind. Electron.*, vol. 45, no. 1, pp. 185–187, Feb. 1998.
[30] N. Diolaiti, G. Niemeyer, F. Barbagli, and J. K. Salisbury, Jr., "Stability of haptic rendering: Discretization, quantization, time delay, and Coulomb effects," *IEEE Trans. Robot.*, vol. 22, no. 2, pp. 256–268, Apr. 2006.
[31] S. Dasgupta and Y. F. Huang, "Asymptotically convergent modified recursive least-squares with data-dependent updating and forgetting factor for systems with bounded noise," *IEEE Trans. Inf. Theory*, vol. IT-33, no. 3, pp. 383–392, May 1987.
[32] M. Jin, S. H. Kang, and P. H. Chang, "Robust compliant motion control of robot with nonlinear friction using time-delay estimation," *IEEE Trans. Ind. Electron.*, vol. 55, no. 1, pp. 258–269, Jan. 2008.
[33] A. Savitzky and M. J. E. Golay, "Smoothing and differentiation of data by simplified least squares procedures," *Anal. Chem.*, vol. 36, pp. 1627–1639, Jul. 1964.
[34] H. K. Khalil, *Nonlinear Systems*, 2nd ed. Englewood Cliffs, NJ: Prentice-Hall, 1996.

Sang Hoon Kang was born in Seoul, Korea, in 1978. He received the B.S. and M.S. degrees in mechanical engineering in 2000 and 2002, respectively, from Korea Advanced Institute of Science and Technology (KAIST), Daejeon, Korea, where he is currently working toward the Ph.D. degree in mechanical engineering.

He is currently with the Robotics and Control Laboratory, KAIST, where he was a Research Assistant from 2001 to 2008, and was involved in three research projects in the field of robotics. His current research interests include robust impedance control of nonlinear plants as applied to robot manipulators and human arm impedance measuring, design and control of assistive robots and rehabilitation robots, and robust control of nonlinear systems, including robots.

Mr. Kang is a Student Member of The Korean Society of Mechanical Engineers (KSME).

Maolin Jin (S'06–M'08) received the B.S. degree in materials science and mechanical engineering from Yanbian University of Science and Technology, Jilin, China, in 1999, and the M.S. and Ph.D. degrees in mechanical engineering from Korea Advanced Institute of Science and Technology (KAIST), Daejeon, Korea, in 2004 and 2008, respectively.

He is currently a Postdoctoral Researcher with the Mechanical Engineering Research Institute, KAIST. His current research interests include robust control of nonlinear plants, time-delay control, robot motion control, and fire-fighting robots.

Dr. Jin is a member of the Institute of Control, Robotics and Systems (ICROS), Korea.

Pyung Hun Chang (S'86–M'89) was born in Pusan, Korea, in 1951. He received the B.S. and M.S. degrees from Seoul National University (SNU), Seoul, Korea, in 1974 and 1977, respectively, and the Ph.D. degree from Massachusetts Institute of Technology (MIT), Cambridge, in 1987, all in mechanical engineering.

From 1984 to 1987, he was involved in a research project in the field of robotics as a Research Assistant at the Artificial Intelligence Laboratory, MIT. Since 1987, he has been on the faculty of the Department of Mechanical Engineering, Korea Advanced Institute of Science and Technology (KAIST), Daejeon, Korea, where he is currently a Professor. His current research interests include robust impedance control of nonlinear plants as applied to robot manipulators and human impedance measuring, dynamic control of redundant manipulators, and target-oriented design approach to rehabilitation robots and safety.

Prof. Chang is a member of the American Society of Mechanical Engineers (ASME), The Korean Society of Mechanical Engineers (KSME), and the Korean Society for Precision Engineering (KSPE).